

Smart Guard: An Intelligent Child Safety and Monitoring System



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AUTHOR'S DECLARATION

We, Mohamed Alaraibi, Reem Madan, Sayed Qasim, and Rahim Alrahim declare that we are the sole authors of the thesis. All sources are appropriately cited. It is further declared that we have fulfilled all the requirements in line with the Quality Assurance guidelines of the Bahrain Polytechnic.

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Abstract

SmartGuard is an intelligent child-safety and monitoring system developed to address recurring incidents of children being forgotten inside school and kindergarten buses a significant concern in Bahrain and across the Gulf region. The system integrates artificial intelligence, multi-sensor monitoring, and automated safety procedures to provide a reliable, real-time method for detecting children inside a bus after it has been turned off. Designed as a proactive solution to prevent suffocation incidents, SmartGuard combines advanced sensing technologies with a modern communication and control framework to support fast, accurate, and automated decision-making.

The SmartGuard system employs a suite of sensors, including mm Wave human-presence detection, thermal imaging, temperature-humidity monitoring, IR-based passenger counting, load sensing, and GPS tracking, to continuously assess occupancy and environmental conditions. These sensors connect to an Arduino MKR NB 1500 microcontroller, which processes incoming data and communicates wirelessly through LTE-NB IoT technology. To enhance detection reliability, the system incorporates a Raspberry Pi-based vision subsystem equipped with dual cameras for live streaming, real-time human detection, and facial recognition to automate student counting. All collected data are prepared for transmission to a cloud-supported monitoring interface for use by parents, supervisors, and administrators.

When a child is detected inside the bus during shutdown conditions, SmartGuard activates a series of immediate safety actions, including alerts, warnings, and automated responses designed to minimize human error and ensure rapid intervention. A functional prototype integrating sensors, LEDs, and vision systems demonstrates accurate detection performance, stable sensor integration, and dependable low-latency response behavior.

SmartGuard represents a significant advancement in child-safety technology by shifting bus monitoring from manual, reactive procedures to a fully automated, proactive, and scalable solution tailored to the region's safety needs.

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“If you are grateful, I will surely increase you.” — Qur’an 14:7

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Dedication

This report is dedicated to our parents and families, whose unwavering support, encouragement, and prayers have guided us throughout our academic journey. Their sacrifices and belief in our potential have been the foundation of our achievements.

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List of Acronyms

WHO	World Health Organization
UNICEF	United Nations Children's Fund
GCC	Gulf Cooperation Council
NSC	National Safety Council
UAE	United Arab Emirates
AI	Artificial Intelligence
IoT	Internet of Things
SDG	Sustainable Development Goal
NHTSA	National Highway Traffic Safety Administration
ROA	Rear Occupant Alert
PCB	Printed Circuit Board
GPIO	General Purpose Input/Output
UART	Universal Asynchronous Receiver-Transmitter
GPS	Global Positioning System
IR	Infrared
LTE	Long-Term Evolution
SMS	Short Message Service

1. INTRODUCTION

1.1 Project Background

Globally, there is a great deal of worry about the safety of kids on school transportation, especially in regions with high temperatures, high levels of humidity, and a significant dependence on bus-based transportation. The World Health Organization (WHO) and UNICEF, among other international health and safety organizations, have repeatedly emphasized the grave dangers of leaving kids inside cars, particularly after they have been turned off, since internal temperatures and humidity can rise quickly and result in potentially fatal circumstances. Due to their inability to control body temperature, their reliance on adult supervision, and their inability to act effectively in emergency circumstances, children are more susceptible.

In recent years, advances in embedded systems, artificial intelligence, and Internet of Things technologies have enabled the development of engineering solutions that have encompassed all fields, particularly smart monitoring solutions capable of addressing safety challenges in real time. These technologies enable the monitoring of environmental conditions, the detection of human presence, the direct transmission of data, and the activation of automated responses without relying entirely on human intervention. In the context of school transport, these systems provide an opportunity and a solution to significantly reduce the risk of accidents caused by human error.

The SmartGuard project was developed in response to these global safety concerns, with a specific focus on enhancing child protection in school and kindergarten buses through intelligent monitoring, automated detection, and proactive safety actions, thereby ensuring a safe and seamless transportation experience for children while providing reassurance and peace of mind to parents and guardians.

1.1.1 History

Historically, school bus safety systems have focused primarily on external factors such as vehicle maintenance, driver training, and compliance with traffic regulations. Although these precautions are crucial for overall transportation safety, internal safety monitoring has historically relied on manual inspections carried out by drivers or supervisors, especially after the bus engine has been turned off. Such methods are nonetheless susceptible to human variables including exhaustion, neglect, poor communication, and workload constraints, as several regional and worldwide occurrences have shown.

Within the Gulf Cooperation Council (GCC) region, including Bahrain, several publicly reported incidents over the past decade have highlighted the serious consequences of these limitations in manual supervision within school transportation systems. Multiple regional investigations and official news reports have documented cases in which children were unintentionally left inside school buses or transport vehicles, leading to severe safety outcomes and, in some cases, fatalities (Al-Watan, 2024; Akhbar Al-Khaleej, 2024; Al-Ayam, 2024; Okaz, 2024; Al-Ayam, 2025; Ryan, 2019; Khaleej Times, 2024). These incidents underscore the inherent vulnerability of school transportation systems that depend primarily on human supervision and manual attendance verification.

These risks are significantly intensified by the extreme climatic conditions prevalent across the GCC. Peer-reviewed experimental research has demonstrated that the temperature inside enclosed vehicles can rise rapidly, even under moderate ambient conditions. A widely cited study published in *Pediatrics* showed that interior vehicle temperatures can increase to dangerous levels within minutes, frequently exceeding thresholds associated with severe heat stress, thereby posing a critical risk to unattended children (McLaren et al., 2005). Supporting these findings, data compiled by the National Safety Council indicate that vehicle interiors can reach approximately 47 °C within one hour, with nearly 80 % of this temperature

increase occurring during the first 30 minutes, and that partial window opening does not provide effective protection against heat buildup (National Safety Council [NSC], 2023).

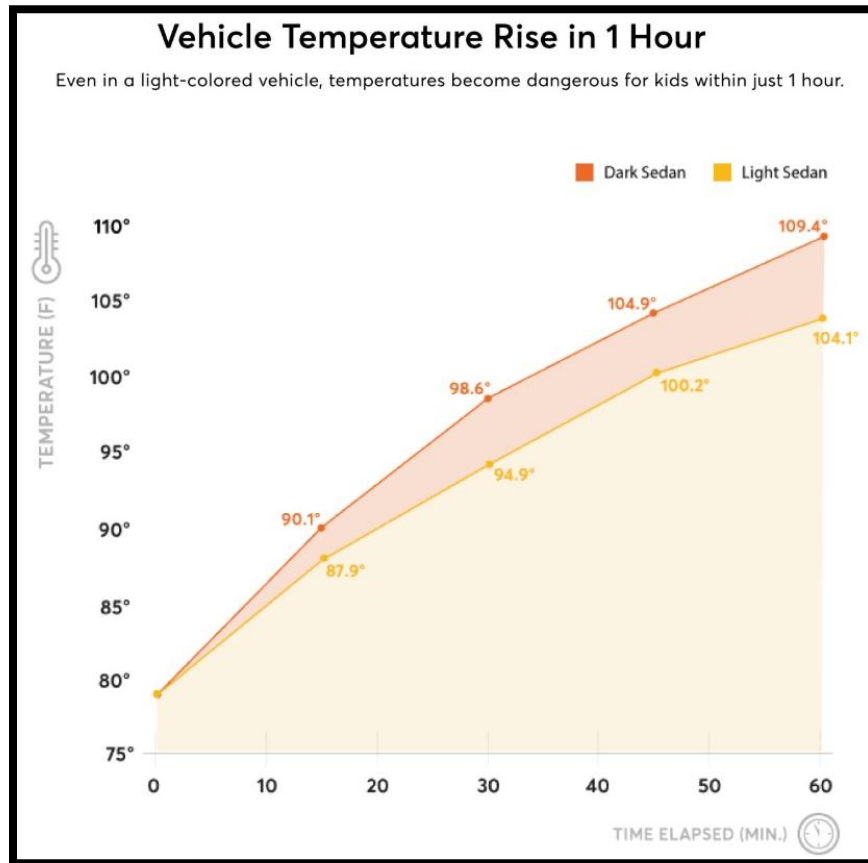


Figure 1: The interior vehicle temperature rise over one hour (Consumer Reports, 2024).

Figure 1 illustrates this phenomenon by presenting measured temperature increases inside light- and dark-colored vehicles over a one-hour period. The figure demonstrates that vehicle interiors can exceed hazardous temperature levels within a short timeframe regardless of vehicle color, confirming that enclosed vehicles can quickly become life-threatening environments for children, particularly in hot climates such as those found in the GCC.

In response to these well-documented risks and repeated regional incidents, authorities and educational institutions have increasingly emphasized the need for technological safety solutions that complement human supervision rather than replace it. The historical evolution of safety systems from manual inspection to sensor-assisted and intelligent monitoring has

laid the foundation for automated solutions such as SmartGuard, which represent a natural progression toward smarter, more reliable, and safer school transportation systems.

1.1.2 Project Idea Generation

The idea for SmartGuard emerged from a combination of real-world observations, academic research, and discussions with professionals and educators. The project team identified recurring reports of children being unintentionally left inside school buses as a critical problem that remained insufficiently addressed by existing solutions. Early brainstorming sessions focused on understanding the root causes of such incidents, including reliance on manual attendance checks, lack of real-time occupancy verification, and absence of automated alerts.

During the idea-generation phase, multiple solution concepts were explored, ranging from simple alarm-based systems to advanced camera-only monitoring solutions. However, further analysis revealed that no single detection method could guarantee reliability under all conditions. This realization led to the concept of a multi-layered system that combines artificial intelligence with multiple complementary sensors to improve accuracy and reduce false alarms.

Consultations with academic supervisors and experienced engineers helped refine the idea into a feasible engineering project that balances innovation, practicality, cost-effectiveness, and scalability. The final concept of SmartGuard was selected due to its strong societal impact, technical relevance, and alignment with modern smart-safety trends.

1.1.3 Overview

SmartGuard is an intelligent child-safety and monitoring system designed to prevent suffocation and safety incidents inside school and kindergarten buses. The system integrates a variety of sensors including mm Wave human detection, thermal imaging, temperature and humidity monitoring, load sensing, infrared passenger counting, and GPS tracking, with a vision-based artificial intelligence subsystem.

A central microcontroller processes sensor data and communicates with a Raspberry Pi based vision unit, enabling real-time human detection, live video streaming, and automated student

counting. When a potential risk is detected, particularly after the bus has been turned off, the system initiates predefined safety actions such as alerts, warnings, and emergency procedures, while simultaneously notifying relevant stakeholders.

The project is implemented as a functional prototype, demonstrating the feasibility and effectiveness of intelligent monitoring systems in improving child safety. SmartGuard is designed to be modular and scalable, allowing for future expansion and potential real-world deployment.

1.1.4 Relevance and Motivation

The motivation behind SmartGuard extends beyond academic requirements and addresses a genuine societal need. In Bahrain and the wider GCC region, recent incidents have underscored the urgency of adopting proactive safety technologies in school transportation. As countries continue to embrace digital transformation and smart infrastructure initiatives, integrating intelligent safety systems into everyday public services has become both practical and necessary.

SmartGuard aligns closely with several United Nations Sustainable Development Goals (SDGs), particularly SDG 3 (Good Health and Well-Being), which emphasizes the prevention of avoidable injuries and fatalities, and SDG 11 (Sustainable Cities and Communities), which promotes safe, inclusive, and resilient urban environments. By reducing the risk of heat-related injuries and preventing critical safety incidents during school transportation, SmartGuard directly supports efforts to protect children's physical well-being and enhance public safety.

Furthermore, the project contributes to SDG 9 (Industry, Innovation, and Infrastructure) by demonstrating the practical application of intelligent sensing, embedded systems, and automated decision-making within real-world transportation systems. Through the integration of smart technologies into school bus operations, SmartGuard offers a scalable, technology-driven solution that enhances trust in school transportation services, reduces reliance on manual supervision, and supports broader national and global sustainability objectives focused on child safety and smart infrastructure development.



Figure 2: Alignment of the SmartGuard System with United Nations Sustainable Development Goals (SDGs).

1.2 Problem Statement

Even with the adoption of standard safety precautions in school transportation systems, human compliance and manual supervision continue to play a major role in present procedures. The majority of school buses rely on drivers or supervisors performing manual attendance checks or visual inspections to confirm the presence of children once the route is completed. Despite the fact that these protocols are designed to guarantee safety, human variables including weariness, disruptions to routines, time constraints, misunderstandings, and cognitive overload might affect them. Therefore, total protection against post-operation safety hazards cannot be ensured by physical supervision alone.

Numerous international and regional incidents have demonstrated that failures in manual supervision can result in children being unintentionally left inside stationary vehicles, including school buses. Public safety reports and transportation safety investigations have identified such incidents as a recurring problem, frequently occurring when established routines are disrupted or when responsibility is assumed rather than verified. Global statistics indicate that a substantial proportion of pediatric vehicular heatstroke fatalities result from children being forgotten inside vehicles, highlighting the critical limitations of relying solely on human memory and procedural checks.

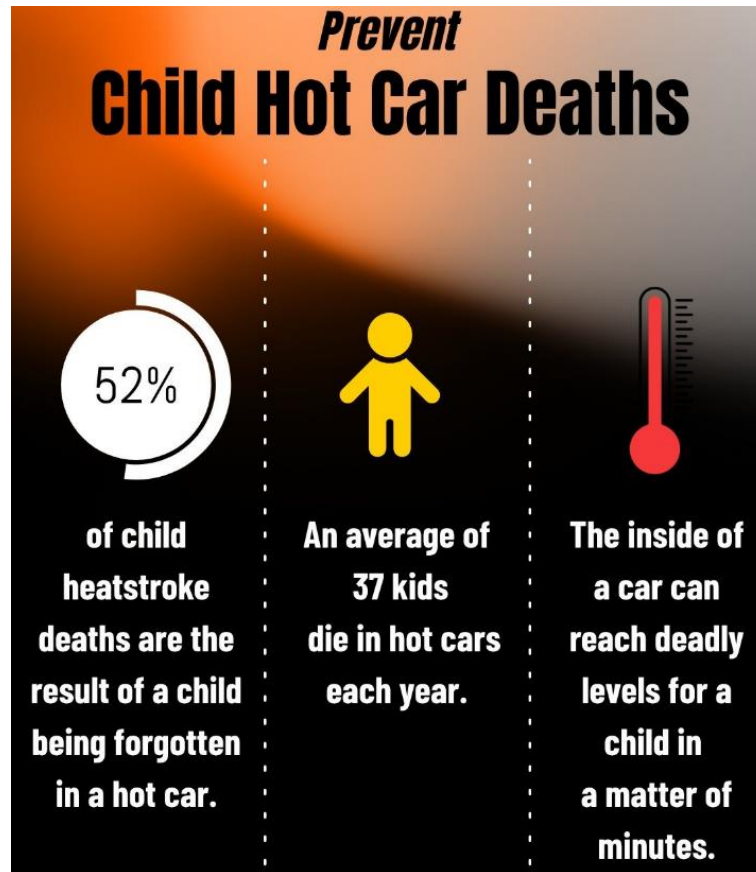


Figure 3: Statistics highlighting the risk of child heatstroke deaths in vehicles

These incidents become particularly dangerous in regions characterized by extreme climatic conditions, such as Bahrain and the wider Gulf Cooperation Council (GCC). Scientific studies and safety authority data have demonstrated that the interior temperature of enclosed vehicles can rise rapidly within a short period of time, even under moderate ambient conditions. Experimental measurements indicate that vehicle interiors can reach life-threatening temperature levels within minutes, creating an environment that poses a severe risk to unattended children, whose bodies heat up significantly faster than those of adults. Existing school bus safety systems primarily focus on external operational factors, including vehicle maintenance, driver training, and compliance with traffic regulations. While these measures are essential for overall transportation safety, they provide limited protection against internal post-operation hazards once the vehicle is parked and unattended. Moreover, most conventional systems lack continuous internal monitoring, real-time child

presence detection, automated alert mechanisms, and immediate response capabilities capable of identifying and mitigating dangerous situations without human intervention. International vehicle safety policy discussions increasingly emphasize the need to complement human supervision with intelligent and automated safety technologies to address these shortcomings.

As a result, a critical gap exists between current school transportation safety practices and the level of protection required to ensure child safety under real-world operating conditions.

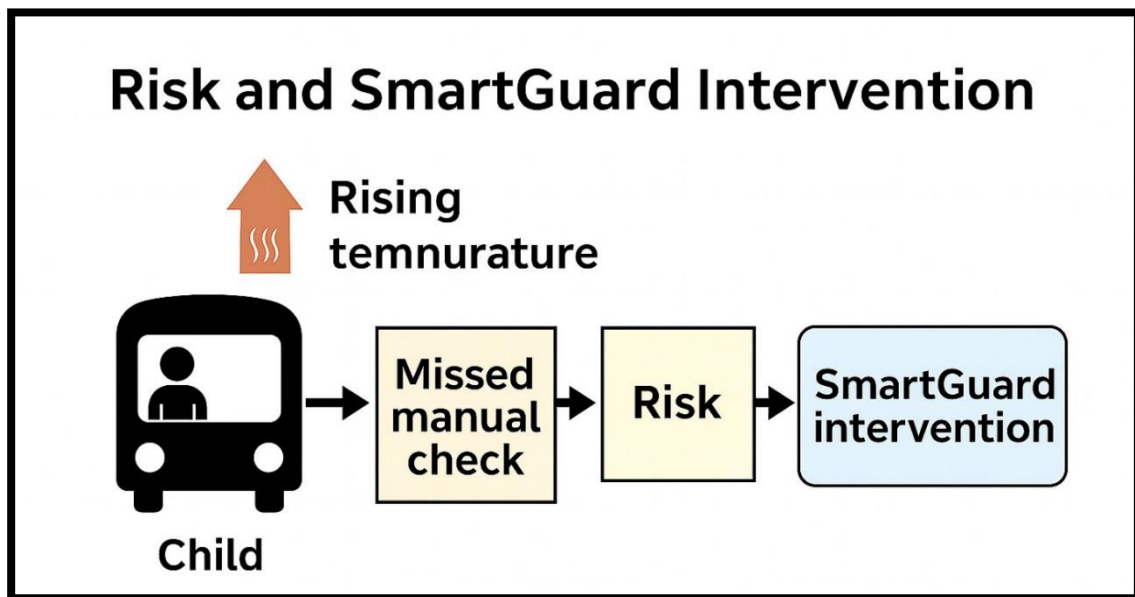


Figure 4: Risk and SmartGuard intervention mechanism.

Addressing this gap forms the core motivation behind the development of the **SmartGuard** system, which aims to enhance child safety in school transportation by reducing dependence on manual supervision and introducing continuous, intelligent monitoring and automated response mechanisms.

1.3 Scope

Several scope definition and prioritizing choices were taken throughout the early planning and design phases of the SmartGuard project to guarantee that the suggested solution addresses the primary safety issue while staying feasible within the academic schedule. With a key focus on kid safety, system dependability, and real-time monitoring, the scope was

established through internal team discussions and supervisory guidance. Non-essential deployment and regulatory tasks were purposefully left out, and activities were ranked according to their safety criticality, functional impact, and implementation practicality. The project's in-scope and out-of-scope tasks and deliverables are compiled in the following tables. This approach supports Bahrain's national commitment to the United Nations Sustainable Development Goals, particularly SDG 3 (Good Health and Well-Being) and SDG 11 (Sustainable Cities and Communities), by enhancing child safety in public transportation systems and promoting safer, more sustainable communities.

Table 1: Project In-Scope Activities / Deliverables

Priority (1–5)	In-Scope Activity / Deliverable	Description
1	Intelligent Child Presence Detection	Design and implementation of a multi-sensor system that uses mm Wave radar and vision-based identification to identify children within a school or kindergarten bus following engine cutoff.
1	Automated Safety Response Mechanism	creation of pre-programmed automated safety measures, such as opening doors and windows, flashing headlights, turning on the horn, and generating alerts, that are activated when a child is detected.
1	Real-Time Alert and Communication System	Using an Arduino MKR NB 1500, LTE-based connectivity is implemented to provide drivers, parents, and school officials with instant alerts.
2	Multi-Sensor Integration	including load sensors for seat occupancy detection, heat sensing to improve detection reliability, infrared sensors for passenger counting, and environmental sensors (temperature, humidity, oxygen).

2	Vision-Based Monitoring and Live Streaming	installation of a Raspberry Pi camera system for live video streaming from inside the bus, facial recognition-based passenger counts, and human detection.
2	Location Tracking System	incorporating a GPS module to deliver real-time bus position data in an emergency.
3	Dashboard / Monitoring Interface	creation of a dashboard or mobile interface that shows the location of the bus, the number of passengers, the weather, and a live video feed.
4	Prototype Development and Testing	building a working prototype that mimics actual bus circumstances and using controlled testing scenarios to validate system performance.

Table 2: Project Out-of-Scope Activities / Deliverables

Out-of-Scope Activity / Deliverable	Description
Commercial Manufacturing	This project does not cover industrial manufacturing, mass production, or system supply-chain optimization.
Regulatory Approval and Certification	Due to time and procedural limitations, legal certification and approval from governmental, safety, or transportation authorities are not included.
Long-Term Field Deployment	The current prototype-based effort does not include long-term real-world testing on functioning school buses.
National Emergency System Integration	The current phase does not include integration with government platforms, traffic control systems, or national emergency services.
Public Mobile Application Deployment	Full deployment of mobile applications on public app stores, including long-term maintenance and updates, is not covered in this project.

1.4 Objectives

The SmartGuard project seeks to solve a practical safety issue pertaining to school and kindergarten transportation by utilizing the information and abilities gained during the Electrical Engineering Technology curriculum. The project's goals are to guarantee that the suggested system successfully improves child safety, lessens the need for manual supervision, and exhibits a dependable engineering solution appropriate for the Bahraini setting. The goals are concentrated on system functioning, detection accuracy, automation, and real-time monitoring.

Table 3: Project Objectives

No.	Project Objective	Description
1	Preventing Child Suffocation Incidents	To avoid heat-related and suffocation mishaps, design a device that can identify whether children are inside a bus after the engine has stopped down.
2	Implement Reliable Child Detection	To increase accuracy and lower false alarms, create a multi-sensor detection system with mmWave radar, thermal sensing, vision-based detection, and auxiliary sensors.
3	Automate Safety Response Actions	When dangerous conditions are identified, automate safety measures such door activation, warning indications, horn activation, and alert triggering.
4	Enable Real-Time Monitoring	Monitor the bus's temperature, humidity, passenger count, and live video stream in real time.
5	Provide Location Tracking	Incorporate a GPS-based tracking system to determine the bus's current location in an emergency.
6	Reduce Human Error	Reduce the need for drivers or supervisors to perform manual inspections by using automated detection, monitoring, and alert systems.
7	Developing a Functional Prototype	Create and test a workable prototype that exhibits the system's dependability and usefulness while simulating actual bus conditions.

8	Ensure Scalability for Future Deployment	Create a scalable system architecture so that it may be expanded, deployed, and integrated into Bahrain's actual school transportation services in the future.
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1.5 Bahrain Perspectives

The environmental, sociological, and legislative aspects of Bahrain's national context make the problem of child safety in school transportation especially important. Bahrain, a nation with a hot desert environment, frequently endures prolonged periods of high ambient temperatures, particularly in the summer when daily highs often surpass 40 °C. These circumstances greatly raise the dangers of enclosed vehicles and intensify the repercussions of any post-route safety supervisory oversight.

Bahrain's worries about the shortcomings of manual supervision in school transportation systems have been heightened by recent events that have been reported this year. According to official local media reports, there have been documented cases in which children were unintentionally left inside school buses after their route had ended, resulting in one death, prompting the competent authorities to conduct investigations and a public discussion on accountability and safety procedures (Al-Ayyam, 2025; Akhbar Al-Khaleej, 2025). These incidents occurred despite the presence of existing procedural safeguards, highlighting the vulnerability of systems that rely primarily on human checks and routine compliance.

Such cases have drawn attention to the fact that even well-established supervision protocols may fail under real operational conditions, including driver fatigue, schedule pressure, and communication breakdowns. In a climate such as Bahrain's, delays in detection can rapidly escalate into life-threatening situations, underscoring the inadequacy of relying solely on manual inspections to ensure child safety.

In parallel with these challenges, relevant authorities in Bahrain conducted workshops in schools to teach children how to handle situations where they are left behind on buses. Bahrain also demonstrated a strong national commitment to digital transformation, smart infrastructure, and technology-enabled public services.

Initiatives related to smart cities, intelligent transportation, and digital governance reflect the country's readiness to adopt advanced technological solutions that enhance safety, efficiency, and parents-trust. Within this framework, the integration of intelligent child safety systems into school transportation aligns with Bahrain's broader vision of proactive risk management and data-driven decision-making.

Accordingly, the adoption of automated, sensor-based safety solutions represents a necessary evolution in school transportation safety within Bahrain. Systems such as SmartGuard address locally observed shortcomings by providing continuous internal monitoring, real-time child presence detection, and automated alert mechanisms that operate independently of human memory or routine adherence. This approach directly responds to recent national incidents while supporting Bahrain's strategic objectives for safer, smarter, and more resilient public services, and aligns with relevant United Nations Sustainable Development Goals related to child safety, innovation, and sustainable urban development.



Figure 5: Kindergarten and school transportation supervision and child safety in Bahrain



Figure 6: Buses safety awareness session conducted in Bahraini schools.



Figure 7: demonstrating emergency response actions inside a school bus.

1.6 Thesis Structure

Table 4: Structure of the thesis and chapter descriptions.

Chapter	Title	Description (According to Rubric)
Chapter 1	Introduction	Presents the background of the project, problem statement, scope, objectives, Bahrain perspectives, and thesis structure. This chapter establishes the context and motivation for the SmartGuard system.
Chapter 2	Literature Review	Reviews relevant academic literature, standards, and existing technologies related to school bus safety, child presence detection, vehicular heat risks, and intelligent monitoring systems. All sources are critically analyzed and properly cited.
Chapter 3	Methodology	Describes the research methods and procedures used in the project, including system design, modeling, simulation, fabrication, data collection methods, and data analysis techniques. It explains how the research was conducted, the rationale behind the chosen methods, and any limitations or challenges encountered.
Chapter 4	Results and Discussion	Presents and discusses all findings generated during the project, including experimental results, system performance evaluation, and any developed products or designs. The discussion explains how the results support the project objectives and their applicability to real-world use in Bahrain.
Chapter 5	Conclusion	Summarizes the main conclusions drawn from the project, highlighting its impacts and benefits for Bahraini society. This chapter also provides recommendations for future work, improvements, and potential extensions of the SmartGuard system.

2. LITERATURE REVIEW

2.1 Heatstroke and Children Forgotten Inside Vehicles

Heatstroke inside vehicles is a documented and frequent cause of serious injury and death among young children. The problem arises when children are left unattended or forgotten inside parked vehicles or school buses after they have been turned off, where the temperature inside the vehicle can rise rapidly to life-threatening levels.

Incident 1: National Heatstroke Cases in the United States

Federal transportation authorities in the United States acknowledge vehicular heatstroke as a recurrent and avoidable cause of child mortality. According to the National Highway Traffic Safety Administration (NHTSA), dozens of children perish every year because of being left inside hot cars; many of these incidents are caused by drivers or caretakers inadvertently forgetting the kid during normal disturbances. Post-trip verification is crucial because, according to NHTSA (National Highway Traffic Safety Administration [NHTSA], n.d.), interior vehicle temperatures can climb dangerously within minutes, even in mild weather. This data demonstrates the limitations of depending just on human recollection and confirms vehicular heatstroke as a systematic safety issue rather than individual irresponsibility.

Incident 2: Physiological and Thermal Mechanisms of Child Heatstroke

Peer-reviewed medical research explains why children are particularly vulnerable when trapped inside enclosed vehicles. Experimental and clinical studies show that vehicle cabins experience rapid temperature increases shortly after engine shutdown, and children's thermoregulatory systems are less capable of dissipating heat compared to adults. As a result, core body temperature can rise quickly, leading to heat exhaustion, heatstroke, organ failure, and death.

The reviewed research confirms that cracking windows or parking in shaded areas does not prevent dangerous temperature escalation, reinforcing the urgency of early detection and intervention (King et al., n.d.).

Incident 3: School Bus Incident in Sharjah, United Arab Emirates (2024)

A four-year-old child was left inside a school bus after dozing off in 2024, according to reports of a major school transportation event in Sharjah, United Arab Emirates. Before being found by workers on a subsequent journey and sent to the hospital, the kid spent more than two hours inside the stopped bus. Despite the child's survival, the event exposed a shortcoming in post-route inspection protocols and highlighted questions about the lack of automated mechanisms that could verify that no passengers were still inside the vehicle (NDTV, 2024). This instance shows that procedural checks may not be enough, even in controlled school transportation systems.

Incident 4: Fatal School Bus Incident in Saudi Arabia (2022)

A fatal case occurred in Saudi Arabia in 2022 when a five-year-old boy died after being left inside a locked school bus. Reports indicated that the driver failed to ensure all children existed before securing the vehicle. Prolonged exposure to extreme heat led to suffocation and death, highlighting the severe consequences of human error in high-temperature environments (Khaleej Times, 2022).

This incident underscores the heightened risk of vehicular heatstroke in hot-climate regions and the need for automated safeguards.

Incident 5: Pediatric Vehicular Heatstroke Fatalities (2000–2016, U.S.)

A large peer-reviewed retrospective study analyzed 541 pediatric heatstroke deaths in the United States from January 1, 2000, to December 31, 2016. The authors found that most deaths occurred when caregivers unintentionally left the child in a parked vehicle. Of the 541 cases, 528 involved single victims and 26 involved two or more fatalities per incident. Most victims were under 5 years old (Guard, 2005).

Incident 6: Egypt-Multiple Children Left in School Buses

In Egypt, a news report detailed multiple cases of children being left behind inside school buses after the end of the school day. The issue involved several incidents where students were forgotten on the bus, sometimes for extended periods, due to lax supervision and procedural failures. In at least three documented cases, children complained of dizziness and exhaustion, likely tied to heat exposure inside the enclosed bus environment. The stories prompted public outcry and calls for stricter safety measures to prevent similar cases in the future, emphasizing that procedural checks alone were insufficient in ensuring child safety during school transport operations (Egelyom, 2025).

Incident 7: Fatal Heatstroke Case in Bahrain (October 13, 2025)

In Bahrain, a tragic vehicular heatstroke death was officially reported by the Northern Governorate Police. On October 13, 2025, a four-and-a-half-year-old boy died after being left inside a closed, unventilated private vehicle for several hours. The boy had been in the car while the driver (a 40-year-old woman) failed to notice him before locking the vehicle and leaving the scene. The Public Prosecution's forensic examination concluded that the cause of death was asphyxiation and severe heat exposure, a result of prolonged confinement in the parked vehicle. The driver was arrested and charged with causing the child's death by negligence and for operating an unlicensed student transport service. This incident highlights that heatstroke fatalities can occur even in regulated environments and underscores the critical need for automated detection systems that do not rely solely on human checks and memory in hot climates like Bahrain's (Police Media Centre, 2025).

2.2 Solutions Implemented Internationally to Minimize Vehicular Heatstroke

Solution 1: National Prevention Programs and Safety Protocols (USA)

In the United States, federal safety agencies promote national prevention strategies that focus on awareness campaigns, driver reminders, and procedural safety checks. The National Highway Traffic Safety Administration (NHTSA) emphasizes practices such as checking rear seating areas after every trip, using reminder systems (e.g., alarms or prompts), and educating caregivers and drivers about the rapid heat rise in parked vehicles. These protocols aim to minimize instances of heatstroke by encouraging behavioral change and consistent adherence to post-trip safety checks, although their effectiveness is limited by dependence on human compliance (National Highway Traffic Safety Administration, n.d.).

Solution 2: Radar-Based Occupancy Detection Systems (Florida, USA)

Certain school districts in the United States have adopted radar-based sensor systems on school buses to detect remaining passengers after the bus route is complete. Radar technology can sense motion and vital signs inside the vehicle even when the child is not moving significantly. When the bus is parked and the engine is switched off, the system scans the interior and alerts the driver if any presence is detected. This reduces reliance on manual checks and provides a technical safety layer that can operate in low visibility conditions (Spectrum News, n.d.).



Figure 8: Radar-based occupancy detection system installed inside a school bus

Solution 3: AI-Based Passenger Tracking (“Safe Bus”) (UAE)

In the United Arab Emirates, a technological solution known as the “Safe Bus” system was developed to address the problem of forgotten children on buses. The system uses artificial intelligence (AI) combined with door-mounted passenger sensors to track boarding and alighting counts. Once the engine is turned off, the system compares the number of passengers who entered versus those who exited. If a mismatch is detected indicating someone may still be inside, it triggers alarms and sends SMS notifications to drivers and administrators. While effective at identifying discrepancies in passenger counts, this approach may produce errors if children enter and exit quickly or in clusters, affecting the count accuracy (Geronimo, 2019).

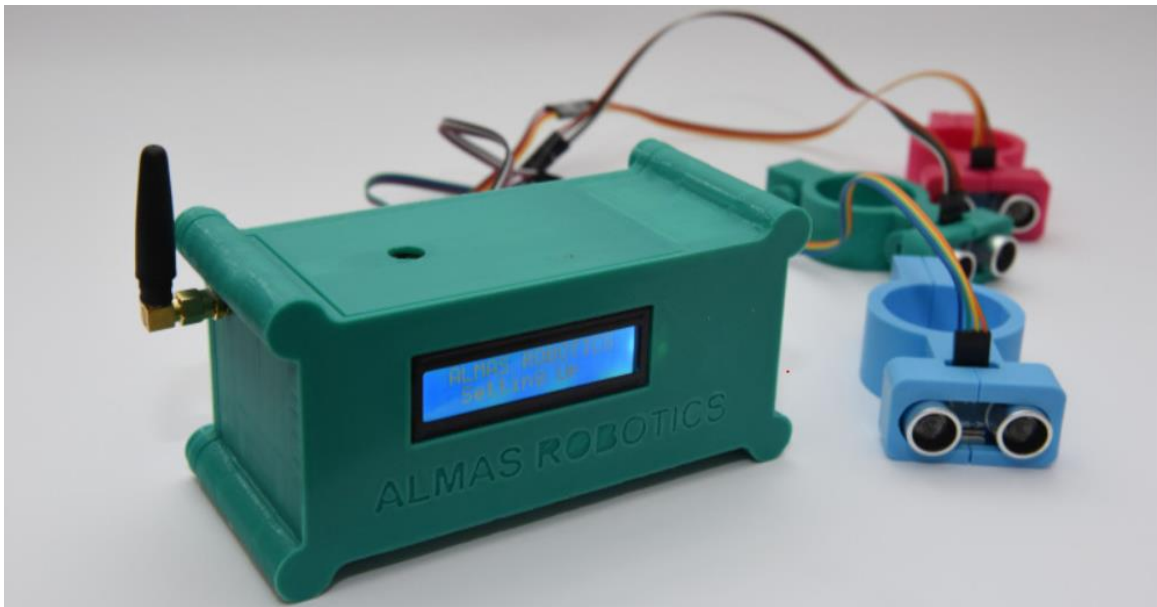


Figure 9: AI-based passenger tracking system (“Safe Bus”) using door-mounted sensors.

Solution 4: Bahrain’s Innovative Seat-Level Alert System (2025)

A local solution from Bahrain involves a patented concept developed by a Bahraini inventor that focuses on direct seat-based detection. The system uses an array of indicator lights, where each light corresponds to a specific seat on a school bus. If a child remains inside when the bus doors are closed, the system triggers an audible alarm and highlights the seat where the child is present. The inventor designed the system to provide seat-specific alerts, helping drivers immediately identify which seat is occupied after shutdown. This innovation

addresses the limitation of generalized detection (presence only) by adding location-specific awareness inside the bus. However, it has not yet been commercialized due to installation cost concerns (estimated at 1800 Bahraini Dinars) and the challenges of securing intellectual property registration without institutional support, though it represents a promising step toward localized preventive technology in regions with high heat exposure and frequent transportation use (Bilad Press, 2025).

Solution 5: Smart School Bus Platforms (Thailand)

Thailand has implemented comprehensive smart school bus systems that integrate multiple technologies, including motion sensors, global positioning system (GPS) tracking, and student monitoring. These platforms connect dashboards and mobile applications used by schools and parents to receive alerts in real time. If a child remains inside the bus after the route ends, notifications are sent promptly. Although this approach increases accountability and transparency, it depends on reliable network connectivity and technically robust integration between devices and platforms (The Voice of Vietnam, n.d.).



Figure 10: Integrated smart school bus platform implemented in Thailand.

Solution 6: Automated Warning Device to Prevent Forgotten Children (Vietnam)

In Vietnam, the Institute of Criminal Science developed an automated warning device specifically designed to prevent children from being forgotten inside school buses. The system is installed within the bus and operates as a post-trip verification mechanism. After the bus route is completed and the engine is turned off, the driver is required to walk to the rear of the bus and press a confirmation button to indicate that the bus is empty. If the driver fails to perform this action within a predefined time window, the system automatically activates a series of safety responses, including audible alarms, flashing lights, and automatic phone calls to emergency contacts.

This multi-layered alert approach is intended to draw immediate attention to the bus and escalate the situation beyond the driver if necessary. The system has been implemented on multiple vehicles and represents a procedural–technological hybrid solution, combining enforced driver behavior with automated alerts. While effective in reducing missed inspections, the system still partially depends on driver compliance, highlighting the continued risk of human error if procedures are bypassed or ignored (Vietnamnet, n.d.).



Figure 11: Automated post-trip warning device installed inside a school bus in Vietnam.

Solution 7: Vehicle Rear Occupant Alert Technology (Automotive Safety - Global Application)

Automotive manufacturers have also contributed to preventing child entrapment through the development of Rear Occupant Alert (ROA) technologies. Companies such as Hyundai Motor Group have introduced radar-based occupant detection systems that monitor the rear seating area of vehicles after the engine is turned off. These systems use radar sensors capable of detecting micro-movements, such as breathing, enabling the detection of occupants even when they are motionless or asleep.

If the driver exits the vehicle while an occupant is still detected in the rear seat, the system issues audible and visual warnings, and in advanced implementations, sends notifications to the driver's smartphone. Although this technology is primarily designed for passenger vehicles rather than school buses, it demonstrates the practical application of radar-based presence detection to address heat-related child fatalities (Hyundai Motor Group, 2023).



Figure 12: Rear Occupant Alert radar-based child presence detection in passenger vehicles.

2.3 Comparison of Applied Solutions

Table 5: Technical Comparison of Existing Solutions

Solution	Country	Core Technology	Automation Level	Detection Reliability
National safety protocols	USA	Procedures, reminders	Low	Low-Medium
Radar occupancy detection	USA	Radar sensors	High	High
AI passenger tracking	UAE	AI + door sensors	Medium	Medium
Smart bus platforms	Thailand	Sensors + GPS + apps	High	Medium-High
Seat-level alert system	Bahrain	Seat indicators + alarms	High	High
Post-trip warning device	Vietnam	Button + alarms	Medium	Medium
Rear occupant alert (ROA)	Global	Radar sensors	High	High

Table 6: Practical Comparison of Existing Solutions

Solution	Human Dependency	Cost Level	Best Suited For	Key Limitation
National safety protocols	High	Low	Awareness & policy	No detection
Radar occupancy detection	Low	Medium-High	Autonomous detection	Calibration
AI passenger tracking	Medium	Medium	Controlled boarding	Counting errors

Smart bus platforms	Low–Medium	Medium-High	Multi-stakeholder use	Network reliance
Seat-level alert system	Low	Medium	Seat-specific checks	Installation cost
Post-trip warning device	Medium–High	Low-Medium	Enforced inspections	Human action
Rear occupant alert (ROA)	Low	Medium	Presence detection	Not bus specific

2.4 Identified Research Gaps and Limitations

The comparative analysis of international solutions presented in Section 2.3 highlights that significant efforts have been made to reduce incidents of children being forgotten inside vehicles and school buses. As summarized in Tables 5 and 6, existing approaches range from procedural safety protocols and awareness campaigns to advanced sensor-based and intelligent monitoring systems. While these solutions demonstrate varying levels of effectiveness, the comparison reveals several recurring limitations that prevent them from fully eliminating the risk of vehicular heatstroke.

One major limitation identified across multiple solutions is the continued dependence on human compliance. Procedural approaches, such as national safety protocols and post-trip verification systems, rely heavily on drivers or caregivers consistently performing manual checks. Even when supported by alarms or reminders, these systems remain vulnerable to memory lapses, fatigue, and routine disruptions. As shown in the comparison, solutions with high human dependency tend to offer lower reliability despite their low cost and ease of implementation.

Technological solutions reduce human reliance but introduce other critical constraints. Counting-based systems, such as AI passenger tracking, may fail when children move in groups or exit rapidly, leading to inaccurate passenger counts. Camera-based detection systems improve situational awareness during loading and unloading but are sensitive to lighting conditions, camera occlusion, and environmental factors. Similarly, smart bus

platforms that integrate GPS, sensors, and mobile applications depend heavily on stable network connectivity and seamless system integration, which may not be consistently available in all operational environments.

Advanced sensor-based solutions, particularly radar and seat-level detection systems, demonstrate higher detection reliability, especially in identifying motionless or sleeping children. However, these systems face challenges related to installation complexity, calibration requirements, and cost, which can limit large-scale deployment. Additionally, many existing solutions rely on a single detection method, increasing the risk of false positives or false negatives if that sensor fails or operates outside optimal conditions. The lack of sensor redundancy remains a notable gap across the reviewed technologies.

Overall, the literature indicates that no existing solution fully balances automation, reliability, cost-effectiveness, and suitability for hot-climate regions. Current systems either rely too heavily on human action, lack robust multi-sensor verification, or impose financial and operational barriers that restrict widespread adoption. These gaps highlight the need for an integrated approach that combines multiple sensing mechanisms, minimizes human dependency, and provides real-time alerts with escalation capabilities. Addressing these limitations forms the basis for the proposed system presented in the following chapter.

2.5 Summary

This chapter reviewed relevant literature related to vehicular heatstroke and the problem of children being forgotten inside vehicles and school buses. Documented international incidents and peer-reviewed studies highlighted the severity of the issue and its recurring nature across different countries and climatic conditions. Existing preventive approaches were examined, including national safety protocols, sensor-based detection systems, artificial intelligence passenger tracking, smart school bus platforms, and automated alert mechanisms. The comparative analysis revealed that while these solutions contribute to risk reduction, they remain limited by factors such as human dependency, single-sensor reliance, operational complexity, and cost constraints. Consequently, the literature indicates a clear

need for an integrated, reliable, and scalable solution capable of minimizing human error while ensuring effective child presence detection in high-risk environments.

In response to these identified gaps, the SmartGuard project is proposed not merely as an academic exercise, but as a practical, low-cost, and deployable smart safety system designed for real-world school transportation environments. The project integrates embedded system development, multi-sensor data acquisition, real-time monitoring, and automated alert mechanisms to address the limitations identified in the literature. Beyond its functional objectives, SmartGuard demonstrates essential engineering competencies relevant to modern intelligent transportation and Internet of Things (IoT) systems.

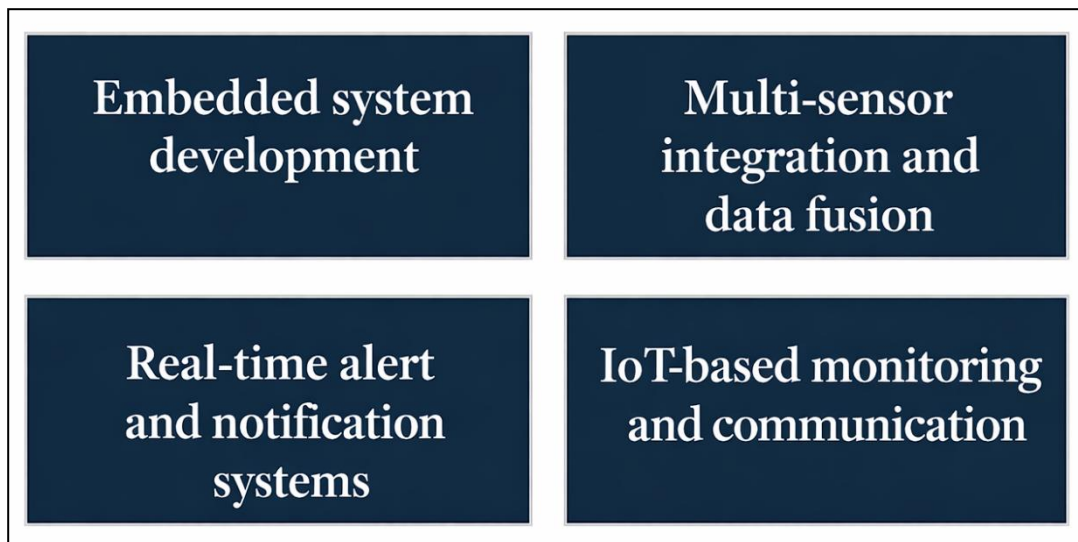


Figure 13: Skills Demonstrated During the SmartGuard Project

3. METHODOLOGY

3.1 Overview

This project adopted an applied engineering methodology centered on the systematic design, development, and integration of a functional safety system prototype. The methodology emphasized practical implementation and verification using real hardware and software components, rather than relying on purely theoretical modeling or simulation-based approaches. This approach was selected to ensure that the SmartGuard system directly addressed the identified safety problem of vehicular heatstroke and suffocation incidents involving children in school buses.

The adopted methodology followed a phase-based development approach, in which the project progressed through a series of clearly defined and logically ordered stages. Each phase was assigned a specific objective, supported by defined procedures, suitable tools and technologies, and measurable outputs. This structured approach enabled traceability across all stages of the project, linking the problem definition to system requirements, design decisions, implementation activities, and planned validation procedures.

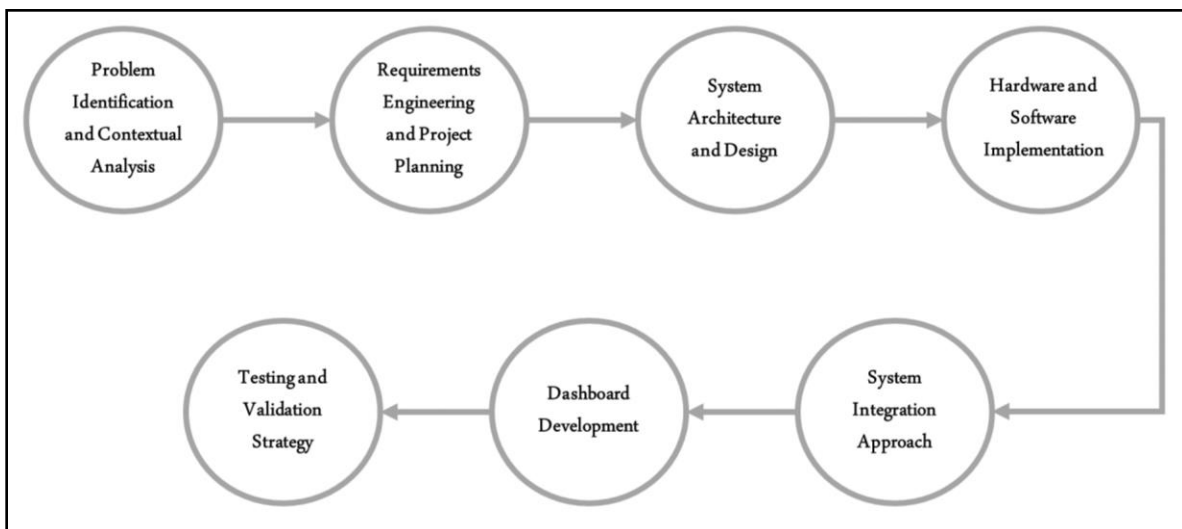


Figure 14: SmartGuard's development phases

The methodology comprised several key phases, beginning with problem identification and contextual analysis to establish the real-world relevance, severity, and underlying causes of vehicular heatstroke incidents in school transportation environments. This was followed by requirements in engineering and project planning, during which functional, non-functional, and safety requirements were systematically derived to guide subsequent design decisions. The next phases focused on system architecture and design, where the overall hardware, software, communication, and actuation structures were defined, followed by hardware and software implementation involving embedded firmware development and vision-based processing. System integration was then conducted to ensure reliable coordination and data exchange between all sensing, processing, communication, and actuation subsystems.

A dedicated dashboard development phase was incorporated to design and implement the system monitoring interface. This phase focused on real-time visualization of sensor data, system status, alerts, and live camera feeds, enabling remote supervision and operational transparency for users and administrators.

Subsequently, a testing and validation strategy phase was defined, outlining verification procedures, validation scenarios, and acceptance criteria used to evaluate system behavior under representative operational conditions. Finally, a comprehensive analysis of constraints, assumptions, limitations, and challenges was conducted to assess the practical feasibility, robustness, and applicability of the proposed SmartGuard solution.

In addition to defining the development phases, the adopted methodology incorporated practical design, fabrication, data collection, and data analysis techniques appropriate for an applied safety-critical engineering project. Hardware fabrication included sensor mounting, physical prototyping, and electronic interfacing, while software development involved embedded firmware programming, computer vision algorithm implementation, and real-time data communication.

Data collection was performed directly from deployed sensors and vision subsystems during controlled experimental scenarios, allowing the system's behavior to be observed under

representative operational conditions. Collected data included sensor detection outputs, environmental readings, system state transitions, alert activation timestamps, and dashboard updates. Data analysis focused on validating logical correctness, detection consistency, timing behavior, and subsystem coordination rather than statistical modeling, reflecting the engineering-oriented nature of the project.

The SmartGuard system was developed using real hardware components, embedded firmware, and vision-processing software. Iterative refinement was employed throughout the development and integration stages to address practical challenges related to sensor behavior, environmental conditions, and hardware-software interaction. This approach enabled early identification of integration issues and informed necessary design adjustments before system-level evaluation.

The adopted methodology is well suited for safety-critical systems, where reliability, redundancy, and verification are essential. By employing a multi-sensor detection strategy, staged safety responses, and continuous system health monitoring, the methodology ensured that the system design and implementation remained aligned with the project objectives and could be evaluated in a structured and transparent manner.

3.2 Project Development Model and Justification

The SmartGuard project was developed using a hybrid V-Model combined with iterative prototyping. This development model was selected to provide a well-structured engineering framework while retaining the flexibility required for practical embedded system development involving hardware and software integration.

The V-Model was chosen due to its emphasis on establishing clear relationships between development phases and corresponding verification and validation activities. For safety-critical applications such as SmartGuard, undetected design errors or integration failures may compromise system reliability and user safety. Consequently, a development model that prioritizes early requirement definition and systematic verification was essential.

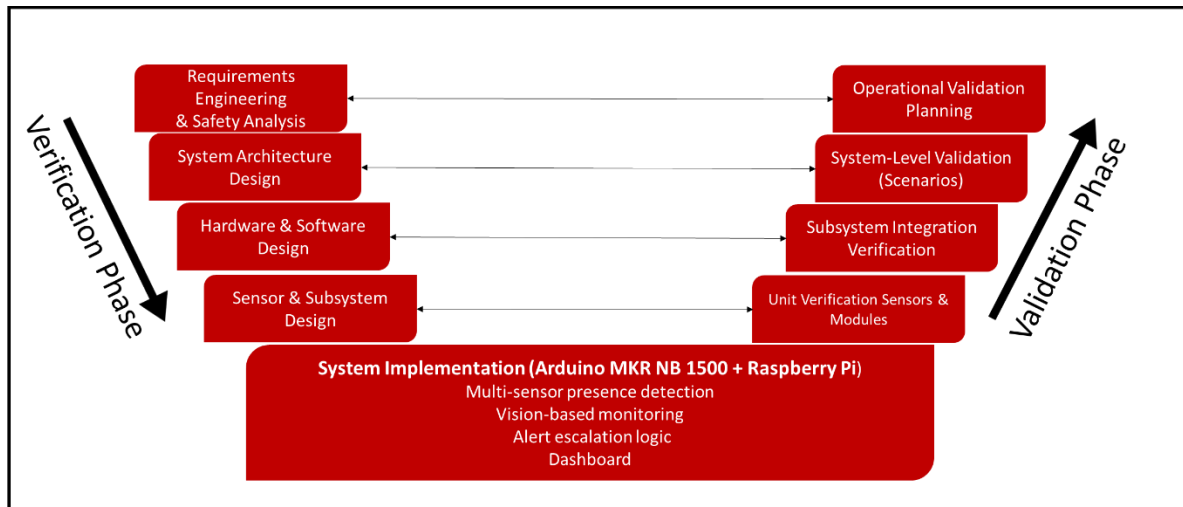


Figure 15: Hybrid V-Model for SmartGuard Development

Within the adopted V-Model, the left-hand side represents system definition and design activities. These activities included problem identification, requirements engineering, and the formulation of the overall system architecture, followed by detailed hardware and software design. At this stage, the functional behavior of sensing, processing, communication, actuation, and monitoring components was defined. Establishing these elements prior to implementation ensured that safety objectives, operational requirements, and system constraints were explicitly specified and remained traceable throughout the project lifecycle.

The implementation phase formed the base of the V-Model and involved the development of embedded firmware for the Arduino MKR NB 1500, vision-processing software for the Raspberry Pi subsystem, and supporting communication logic. System components were implemented incrementally, allowing each module to be functionally verified before progressing to higher levels of integration. This phase focused on achieving reliable low-level operation of sensors, processing units, and alert mechanisms.

In parallel with core system implementation, a dedicated dashboard development activity was incorporated as part of the implementation and integration stages. The dashboard was treated as a system-level interface rather than a standalone application, and its design was driven directly by system requirements. This activity involved defining data visualization requirements, system status indicators, alert presentation logic, and live video streaming

integration. Including dashboard development within the V-Model ensured that user interaction, monitoring capabilities, and operational transparency were validated against system requirements rather than added as an afterthought.

The right-hand side of the V-Model addressed verification and validation planning. Unit-level verification was defined for individual sensors, embedded software modules, and dashboard data handling components. Integration-level verification focused on ensuring correct interaction between sensing, processing, communication, and visualization subsystems. System-level validation was planned to assess overall system behavior, including dashboard accuracy and responsiveness, under representative operational scenarios.

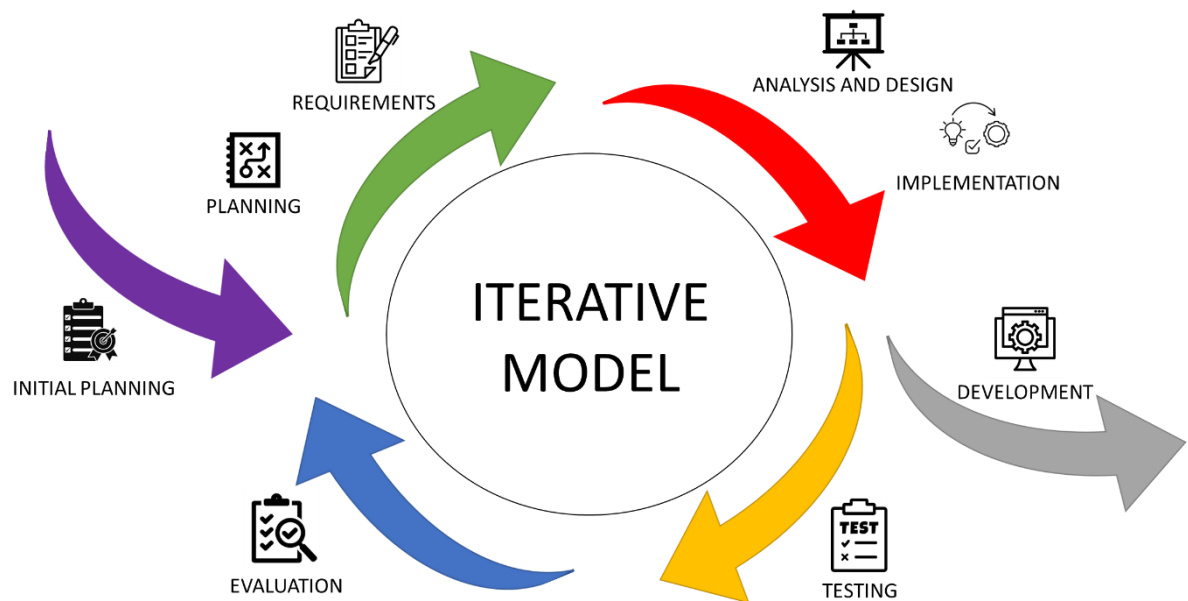


Figure 16: Iterative Prototyping Integrated with the V-Model

To accommodate real-world engineering constraints, iterative prototyping was incorporated across all stages of the V-Model. This iterative approach enabled continuous refinement of both system functionality and dashboard behavior in response to challenges such as sensor calibration variability, environmental effects, communication latency, and hardware–software interaction issues. Feedback from intermediate testing and integration stages

informed design improvements, contributing to enhanced system robustness, usability, and reliability.

The hybrid V-Model with iterative prototyping was selected because it:

- Ensures clear traceability between system requirements, design decisions, dashboard functionality, and validation activities,
- Provides a structured and academically rigorous development framework suitable for safety-critical systems,
- Allows flexibility to refine system components and user-facing interfaces in response to practical implementation challenges.

Overall, this development model was well suited to achieving the objectives of the SmartGuard project, supporting the development of a reliable, transparent, and safety-oriented system through a rigorous and systematic engineering process.

The inclusion of dashboard development within the V-Model was methodologically justified due to its role in system supervision, verification, and operational transparency. The dashboard was treated as a verification and validation interface through which system outputs, sensor states, and alert logic could be observed and evaluated in real time. This ensured that system behavior was not only functionally correct at the embedded level but also interpretable and verifiable by users, thereby strengthening both system validation and usability assessment.

3.3 Problem Identification and Contextual Analysis

The first phase of the SmartGuard project was dedicated to the systematic identification and contextual analysis of the safety problem addressed by the proposed system. The primary objective of this phase was to establish a clear, evidence-based understanding of vehicular heatstroke and suffocation incidents involving children left unattended inside school buses, and to confirm that the problem represents a recurring, high-risk, and safety-critical challenge that warrants an engineering intervention.

3.3.1 Problem Identification Methodology

Problem identification was conducted using a structured, multi-source investigation approach. This involved reviewing documented cases from international safety organizations, governmental and non-governmental reports, academic studies, and verified news articles related to child vehicular heatstroke incidents. Emphasis was placed on incidents occurring in hot-climate regions, including the Gulf region, where ambient temperatures can rise rapidly and significantly increase the risk of fatal outcomes within short time periods.

In parallel with documentary research, exploratory research was conducted to assess the feasibility of potential technological solutions within the scope of a senior engineering project. This activity was followed by focused brainstorming sessions in which multiple solution concepts were proposed, discussed, and critically evaluated based on technical feasibility, innovation potential, safety impact, and implementation constraints.

To strengthen contextual understanding, insights were also gathered through consultations with professionals, including engineers, supervisors, and field experts, as well as academic staff. These discussions provided practical perspectives on existing safety practices, operational challenges in school transportation environments, and limitations of current preventive measures. The combination of documentary research and professional consultation ensured that the problem definition was grounded in both empirical evidence and real-world operational experience.

3.3.2 Contextual Analysis and Root Causes

Contextual analysis revealed that vehicular heatstroke incidents are predominantly driven by human and procedural factors rather than mechanical failure. Common contributing causes identified across multiple cases included driver oversight after route completion, reliance on manual post-trip inspection procedures, children falling asleep and remaining unnoticed, and the absence of automated verification mechanisms to confirm bus occupancy after engine shutdown.

This analysis demonstrated that existing preventive approaches are highly dependent on human memory, routine compliance, and manual checks. Under real operational conditions, particularly during busy schedules, fatigue, or distractions, these practices are prone to failure. Furthermore, contextual review highlighted that many existing technological solutions rely on single detection mechanisms or require manual confirmation, making them vulnerable to false negatives, sensor failure, or missed detections when occupants remain motionless.

The analysis also identified a gap between the severity of the risk and the level of automation currently deployed in school transportation safety systems. Despite the availability of sensing and communication technologies, their integration into autonomous, fail-safe systems remains limited.

3.3.3 Problem Definition and System Need

Based on the findings of the identification and contextual analysis, the core problem was formally defined as the lack of a reliable, autonomous, and multi-layered safety system capable of detecting human presence inside school buses after engine shutdown and initiating timely safety actions without dependence on human intervention.

This definition emphasizes the need for:

- Autonomous operation independent of driver actions.
- Redundant detection mechanisms to reduce single-point failures.
- Immediate and staged safety responses under hazardous conditions.
- Continuous monitoring during post-trip scenarios.

The initial system concept emerging from this phase focused on combining complementary sensing technologies, to improve detection reliability while minimizing false alarms. This concept aligned with regional safety needs, global trends in intelligent transportation systems, and the practical constraints of an academic engineering project.

3.3.4 Phase Outcomes

The outcomes of this phase included:

- A clearly articulated and justified problem statement.
- Identification of key risk factors and failure modes associated with current practices.
- Definition of high-level safety objectives guiding system behavior.
- Selection of an initial system concept suitable for further engineering development.

These outcomes directly informed Phase 2: Requirements Engineering and Project Planning by providing a structured foundation for deriving functional, non-functional, and safety requirements. As such, Phase 1 established the analytical and contextual basis upon which all subsequent design, implementation, and validation activities of the SmartGuard project were built.

3.4 Requirements Engineering and Project Planning

The second phase of the SmartGuard project focused on systematically transforming the identified safety problem into a well-defined set of engineering requirements and a structured project plan. The primary objective of this phase was to clearly specify what the system must achieve, how it should operate, and under which conditions it must function, to guide all subsequent design, implementation, and integration activities in a controlled and traceable manner.

3.4.1 Requirements Engineering Process

Requirements engineering was conducted through an iterative and collaborative process that built directly on the problem definition established in Phase 1. Initially, the project concept was based on a relatively simple detection approach using a single camera and a millimeter-wave human detection sensor. However, through structured discussions, research activities, and critical evaluation, it became evident that a more comprehensive and resilient solution

was required to adequately address the safety risks associated with children being left inside buses after engine shutdown.

To refine the system requirements, multiple activities were undertaken:

- Review of real-world incident scenarios related to vehicular heatstroke and delayed emergency response.
- Brainstorming sessions among team members to explore detection strategies, failure cases, and improvement opportunities.
- Consultations with academic supervisors, university professors, and experienced professionals to validate technical feasibility and practical relevance.
- Analysis of environmental and operational conditions specific to school bus usage in hot climates.

These activities enabled the identification of system behaviors required not only for detection, but also for prevention, escalation, and remote monitoring.

3.4.2 Functional Requirements Definition

Based on the above analysis, functional requirements were defined to describe the essential capabilities of the SmartGuard system. Core functional requirements included the ability to:

- Detect human presence inside the bus after engine shutdown using multiple sensing mechanisms.
- Monitor environmental conditions such as temperature and humidity that contribute to heat-related risk.
- Maintain awareness of passenger count and seat occupancy to support presence verification.
- Determine the bus location to support emergency response and remote supervision.
- Execute automated safety actions when hazardous conditions persist.
- Provide real-time system monitoring and alerts through a dashboard accessible to authorized users.

As the project progressed, additional functional requirements were introduced to enhance reliability and reduce single-point failure risks. These included integration infrared sensors for passenger counting, load sensors for seat occupancy detection, GPS for location tracking, and a driver-facing display to present key safety information during operation. The requirement for a mobile or web-based dashboard was also formalized to allow parents and school administrators to remotely monitor system status and receive alerts.

3.4.3 Non-Functional and Safety Requirements

In parallel with functional requirements, non-functional requirements were defined to capture the expected performance and operational characteristics of the system. These requirements emphasized reliability, low response latency, continuous operation, and robustness under high-temperature and vibration-prone environments typical of school buses.

Given the safety-critical nature of the application, safety requirements were explicitly defined as a separate category. These requirements governed how and when safety actions could be triggered. To prevent false alarms or unintended activations, the system was required to apply multi-sensor confirmation rules rather than relying on a single detection source. Time-based escalation logic was also specified to ensure that safety responses intensified appropriately if hazardous conditions remained unresolved.

Fail-safe behavior was considered during requirements definition to address potential sensor failure, communication disruption, or unexpected system states. This ensured that local safety actions could still be executed even if external communication or dashboard connectivity was temporarily unavailable.

3.4.4 Project Planning Activities

Following requirements definition, structured project planning activities were conducted to organize development tasks and establish a clear implementation roadmap. The planning process focused on aligning system requirements with practical development stages and available project resources.

Key planning activities included:

- Dividing the project practical implementation into logical phases such as component selection, software development, hardware development, integration, testing, and finalization.
- Defining task dependencies between sensing, embedded software, vision processing, communication, and dashboard development.
- Developing an initial Gantt chart to illustrate the project timeline, task durations, and milestone sequencing.
- Distributing responsibilities among team members to ensure parallel progress across hardware, software, and documentation tasks.

This planning stage also involved preparing essential project documentation, including the project proposal, system concept diagrams, and preliminary dashboard layouts. These planning artifacts provided a structured reference for tracking progress and managing iterative changes during later phases.

3.4.5 Phase Outputs and Transition to Design

The outputs of this phase included a comprehensive set of functional, non-functional, and safety requirements, along with a structured development plan defining implementation priorities and phase transitions. These outputs established a clear link between the identified safety problem and the technical solution to be developed.

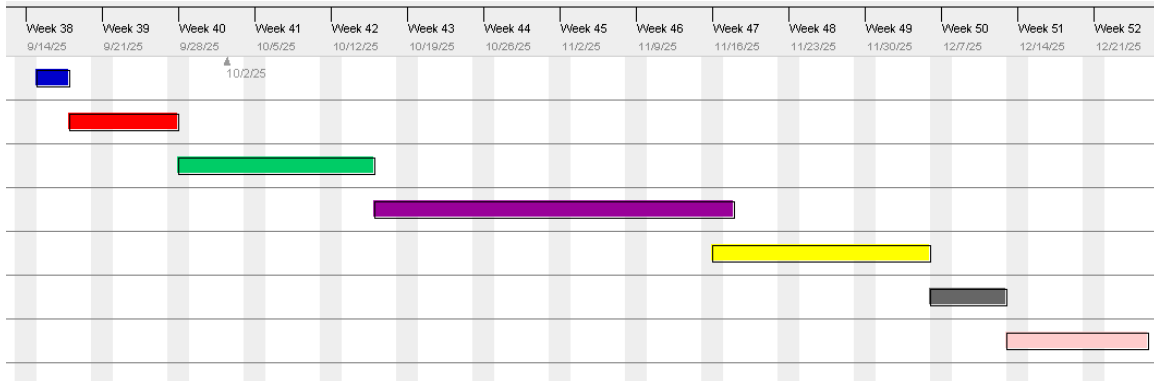


Figure 17: SmartGuard Gantt Chart

Name		Begin date	End date	Duration
Identifying The Problem	●	9/15/25	9/17/25	3
Research and Planning	●	9/18/25	9/27/25	10
Select the Components	●	9/28/25	10/15/25	18
Software Development	●	10/16/25	11/17/25	33
Hardware Development	●	11/16/25	12/5/25	20
Integration and Testing	●	12/6/25	12/12/25	7
Finalization, Report, and Presentation	●	12/13/25	12/25/25	13

Figure 18: SmartGuard Gantt Chart Details

By completing this phase, the project moved from a conceptual idea into a well-defined engineering problem with explicit objectives, constraints, and implementation pathways. This provided a solid foundation for Phase 3: System Architecture and Design, ensuring that all architectural decisions were directly traceable to validated requirements and planned development activities.

3.5 System Architecture and Design

The third phase of the SmartGuard project focused on defining the overall system architecture and detailed design of the hardware, software, communication, and actuation subsystems. The primary objective of this phase was to translate the requirements established in Phase 2 into a robust, modular, and scalable architectural framework capable of supporting reliable human presence detection, environmental monitoring, real-time communication, and autonomous safety responses within a safety-critical school transportation environment.

System architecture development was driven directly by the functional, non-functional, and safety requirements identified earlier. A layered and modular design approach was adopted to ensure a clear separation of responsibilities among system components, simplify system integration, and support future scalability. This architectural strategy allowed individual subsystems to operate independently while enabling coordinated system-level decision-making when safety-critical conditions were detected.

At the highest level, the SmartGuard system architecture was structured into four primary layers: sensing, processing, communication, and actuation. The sensing layer was responsible for acquiring real-time data related to human presence and environmental conditions inside the bus. This layer integrated multiple complementary sensors, including millimeter-wave radar for presence detection, thermal imaging for heat pattern analysis, load sensors for seat occupancy detection, infrared sensors for passenger counting, and temperature and humidity sensors for environmental monitoring. The use of multiple sensing modalities was intentionally designed to provide redundancy, increase detection reliability, and reduce dependence on any single sensor.

The processing layer was designed around two computing platforms with clearly defined and complementary roles. The Arduino MKR NB 1500 microcontroller acted as the primary control unit, handling sensor data acquisition, decision-making logic, system state management, and actuation control. In parallel, a Raspberry Pi subsystem was incorporated to manage computationally intensive tasks, including vision-based monitoring, facial recognition, motion detection, and live video streaming. This division of processing

responsibilities ensured efficient resource utilization while maintaining real-time system responsiveness.

The communication architecture was designed to support both internal and external data exchange. Internally, structured communication pathways enable reliable data transfer between sensors, processing units, and actuation components. Externally, cellular connectivity was integrated to allow real-time transmission of system data, alerts, and location information to the remote monitoring dashboard. The communication design prioritized reliability, low latency, and continuous availability to support remote supervision and emergency response requirements.

The actuation subsystem was architected to execute safety actions autonomously once hazardous conditions were confirmed. This subsystem included provisions for triggering audible alarms, visual indicators, automatic door and window opening mechanisms, and external signaling actions. These actuations were tightly coupled with the system's decision logic to ensure deterministic, predictable, and timely responses during emergency scenarios. In addition to functional considerations, system-level safety and reliability features were embedded within the architecture. Initialization mechanisms were designed to verify sensor availability and system readiness during startup, while runtime health monitoring pathways continuously assessed sensor status, communication integrity, and processing performance. These architectural safeguards ensured that system integrity could be monitored throughout operation and that abnormal conditions could be detected and handled appropriately.

The outputs of this phase included a comprehensive system architecture definition, subsystem interaction models, data flow representations, and interface specifications. These outputs served as the technical blueprint for Phase 4: Hardware and Software Implementation, ensuring that subsequent development activities adhered to a well-defined, modular, and safety-oriented architectural foundation.

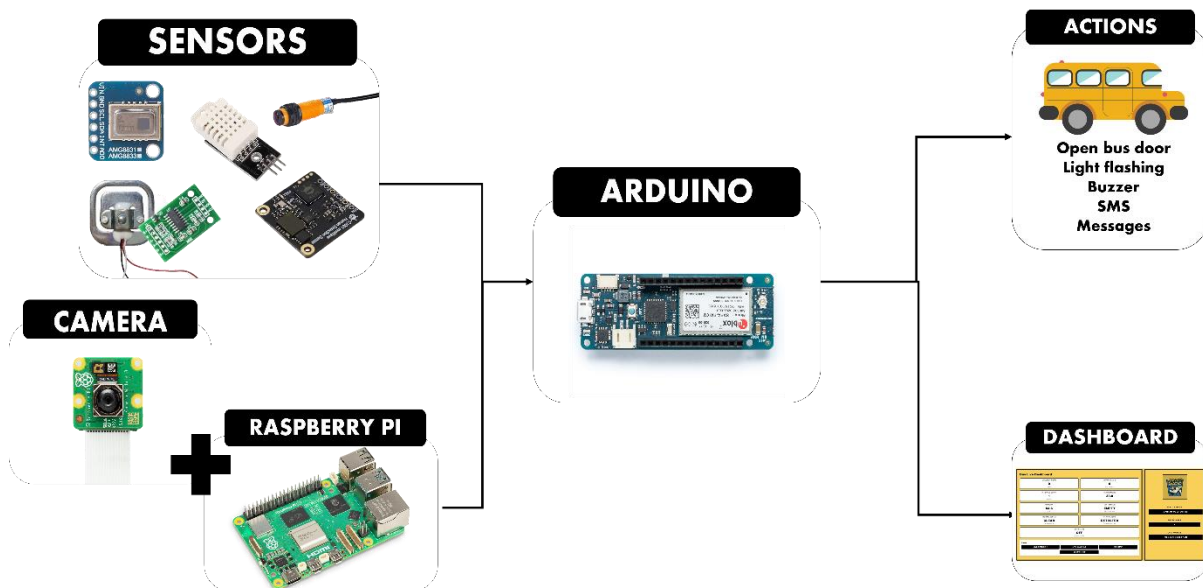


Figure 19: SmartGuard System Fabrication

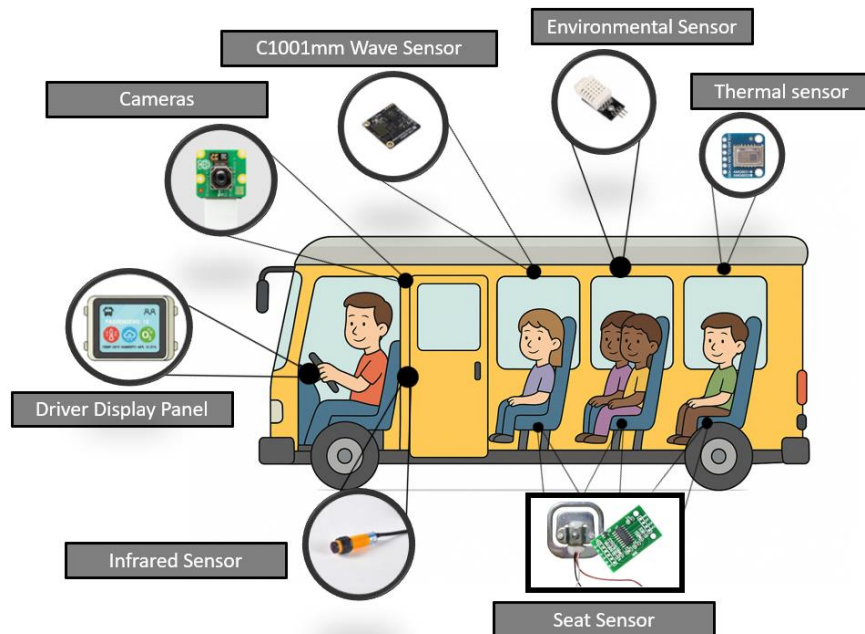


Figure 20: SmartGuard System Diagram

3.6 Hardware and Software Implementation

The fourth phase of the SmartGuard project focused on the detailed implementation of the system architecture defined in Phase 3. The primary objective of this phase was to translate the approved architectural design into a functional prototype through selecting appropriate components aligned with the engineered requirements, implementing the software modules for sensing, decision-making, communication, and monitoring, and developing the physical hardware prototype including PCB planning and enclosure fabrication. The implementation was carried out incrementally to ensure each subsystem was verified at a basic functional level before proceeding to full system integration. This phase represented the transition from conceptual design to practical system realization.

3.6.1 Component Selection Based on Requirements

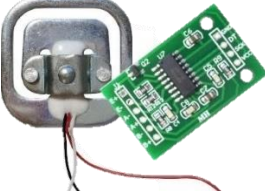
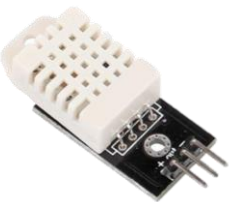
Component selection was performed using the functional, non-functional, and safety requirements defined previously as the primary decision criteria. The selection process aimed to ensure that each requirement was supported by at least one suitable component, while also maintaining redundancy for safety critical detection functions. In addition, practical engineering constraints were considered, such as compatibility with the selected processing platforms, availability in the local market, integration effort, power requirements, physical mounting suitability, and reliability in a high-temperature vehicle environment.

Table 7: Component Selection Criteria Applied in the SmartGuard Project

Selection Criterion	Description	Relevance to SmartGuard
Requirement Coverage	The component must directly satisfy one or more functional, non-functional, or safety requirements.	Ensures all core system functions (human presence detection, environmental monitoring, passenger counting, alert escalation, location tracking, and remote monitoring) are

		fully supported by the selected hardware.
Reliability and Redundancy	Safety critical functions should not rely on a single sensing modality complementary technologies are preferred.	Supports the multi-sensor detection strategy by combining radar, thermal, load-based, infrared, and vision-based sensing to reduce false negatives and improve system reliability.
Platform Compatibility	The component must interface reliably with the selected processing units using standard protocols (UART, I2C, GPIO, etc.).	Ensures seamless integration with the Arduino MKR NB 1500 and Raspberry Pi without requiring custom hardware interfaces or excessive signal conversion.
Environmental Suitability	The component must tolerate high temperatures, vibration, and operational conditions typical of school bus interiors.	Addresses the harsh environmental conditions associated with vehicular deployment, particularly in hot-climate regions.
Integration Complexity	Preference is given to components with stable libraries, clear documentation, and manageable calibration and configuration procedures.	Reduces development time, integration risk, and debugging complexity during software and hardware implementation phases.
Cost and Availability	The component must be affordable and reasonably available within the constraints of an academic prototype.	Ensure project feasibility within time and budget limitations while allowing for replacement or scaling if required.

Table 8: SmartGuard Hardware Components and Functions

Component	Image	Function
Arduino MKR NB 1500		Main control unit: sensor acquisition, decision logic, actuation triggers, system state management
Raspberry Pi		Vision processing, streaming, and supports dashboard server logic
C1001 mm Wave Sensor		Human presence detection
AMG8833 Thermal Sensor		Detects thermal patterns and verifies body heat presence
HX711 Load Cell		Seat occupancy sensing
IR Sensors (E18-D80NK)		Passenger entry/exit counting
DHT22		Bus temperature/humidity monitoring

Cameras		Face-recognition, motion detection, and live stream
GPS Module		Location tracking

3.6.2 Software Implementation

Software implementation focused on developing a reliable, fault-aware control system capable of acquiring multi-sensor data, managing system states, evaluating hazardous conditions, and executing safety logic in a deterministic manner. Given the safety-critical nature of the SmartGuard system, software development emphasized predictability, modularity, and traceability rather than computational optimization or performance tuning. The implementation followed a progressive and incremental strategy, beginning with isolated sensor handling and advancing toward coordinated system-level control. This ensured that each software component could be validated independently before participating in safety decision-making.

Software architecture and code organization

At the architectural level, the software was divided into two cooperative layers:

1. **Embedded Control Layer (Arduino MKR NB 1500):** This layer served as the core of the SmartGuard system. It was responsible for deterministic real-time tasks, including sensor data acquisition, system state transitions, multi-sensor validation logic, and actuation control. All safety-critical decisions were executed within this layer to ensure predictable timing and behavior.

2. **Vision Processing Layer (Raspberry Pi):** This layer handled computationally intensive tasks such as camera acquisition, image processing, motion detection, and human presence analysis.

A modular coding structure was adopted across both layers. Each sensor, subsystem, and functional block was implemented as an independent module with dedicated initialization, update, and error-handling routines. This modularity simplified debugging, allowed isolated testing, and supported incremental system expansion.

Sensor driver development and verification procedure

Sensor software development was performed using a standardized, repeatable procedure to ensure consistency across different sensing modalities. Each sensor was implemented and verified independently before being allowed to influence system-level logic.

The development process began with low-level interface configuration. Communication protocols such as GPIO, I2C, and UART were configured according to sensor specifications, with explicit handling for timing constraints and signal polarity. Once communication was established, raw sensor readings were logged over time to verify signal stability and detect communication anomalies.

Calibration and threshold definition were then performed where required. For example, load cell baselines were established under no-load conditions to compensate for mechanical offsets, while mm Wave radar sensitivity thresholds were adjusted to reliably detect low-movement presence without triggering false detections. Infrared sensors were verified for consistently triggering distances to ensure reliable passenger entry and exit detection.

To support fault awareness, software-based health-monitoring logic was incorporated for each sensor. This logic monitored communication availability, value plausibility, and update frequency. Sensors that failed health checks were flagged as unavailable and excluded from safety decision-making. A dedicated sensor status interface was implemented within the dashboard to visually indicate the operational health of each sensing component in real time.

Time-controlled sampling mechanisms were used to ensure that sensor readings were refreshed at appropriate intervals without blocking other system tasks. This prevented timing conflicts and supported stable real-time operation.

← Back to Dashboard

Bus Sensors Status	
C1001	OK
LOAD	OK
DHT	OK
IR	OK
THERMAL	OK
GPS	OK

Figure 21: Sensor status monitoring interface indicating real-time health and availability of SmartGuard components.

Multi-sensor data fusion and system state management

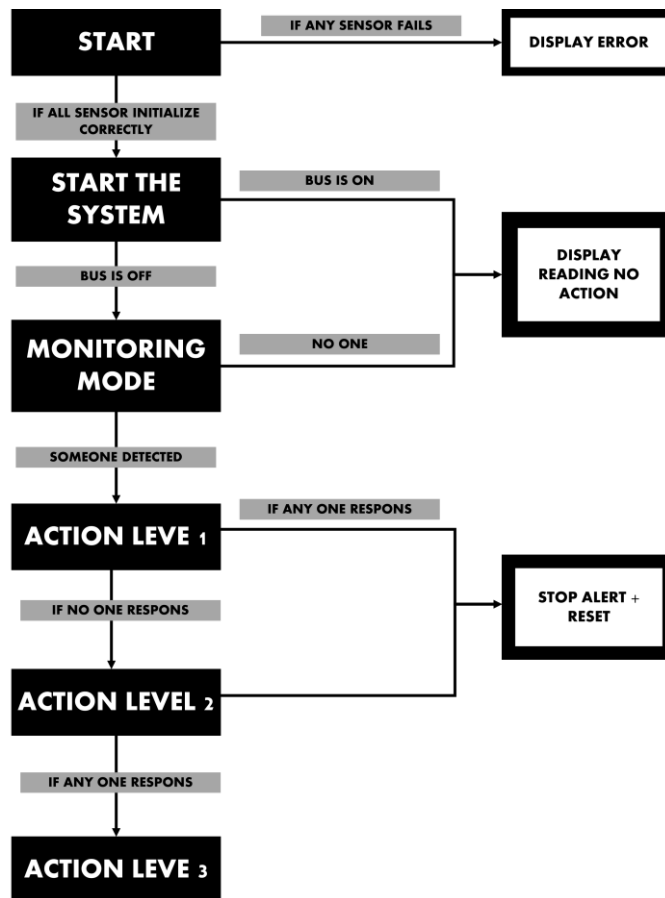


Figure 22: SmartGuard Flowchart

Following individual sensor verification, a unified decision-making layer was implemented using a structured system state machine. This approach allowed the SmartGuard system to transition between operational modes in a controlled and traceable manner.

System states were defined to represent distinct operating conditions, such as normal monitoring during engine operation and enhanced safety verification following engine shutdown. Transitions between states were governed by explicit conditions derived from sensor inputs and system timing.

Multi-sensor data fusion was implemented using confirmation logic rather than numerical averaging. Presence detection decisions required corroboration from multiple independent sensing layers, such as radar, thermal, load, or passenger count data. This reduced the likelihood of false positives or false negatives caused by individual sensor anomalies.

Fail-safe behavior was defined for scenarios involving sensor malfunction, ensuring that unreliable inputs did not compromise system safety.

Alert control and escalation logic

The SmartGuard system implements a structured, time-based alert escalation strategy to ensure timely intervention while minimizing unnecessary or premature emergency actions. The escalation logic is activated only after the bus engine is detected as OFF and human presence is confirmed through multi-sensor verification. Each alert stage is associated with predefined actions and time thresholds, ensuring deterministic and traceable system behavior.

The escalation strategy was designed to prioritize human intervention at early stages while progressively increasing the level of response if the hazardous condition persists. This approach aligns with safety-critical system design principles by balancing responsiveness, reliability, and proportionality of actions.

Table 9: Alert Stages and Control Logic in SmartGuard

System State / Alert Stage	Trigger Condition	System Actions	Purpose
Monitoring Mode	Engine ON	System monitors sensors and logs data	Normal operation without safety intervention
Verification Mode	Engine OFF	Multi-sensor confirmation of human presence	Prevents false alerts after trip completion
Alert Stage 1	Any two sensors confirmed after engine OFF	SMS message sent to bus driver	Immediate notification to prompt manual inspection
Alert Stage 2	No response within 2 minutes	Bus door opens automatically, external lights flash, horn sounds, message sent to parents including bus location	Attracts attention, improves ventilation, and informs guardians
Alert Stage 3	No response within 5 minutes	Emergency services are contacted automatically	Ensure external intervention in critical situations
Fault Mode	Sensor or system anomaly detected	Safety actions inhibited or limited, fault reported	Prevents unsafe operation under unreliable conditions

The three-stage escalation model was selected to reflect realistic operational behavior and safety priorities in school transportation environments. The first stage relies on the bus driver as the fastest and most direct responder. If the driver fails to respond within a defined timeframe, the second stage introduces automated physical actions and parental notification to increase visibility, ventilation, and external awareness of the situation.

The final stage escalates the response to emergency services when no corrective action is detected after a prolonged period. This ensures that the system does not rely indefinitely on human intervention and can act autonomously to prevent severe harm.

Time thresholds were selected to allow sufficient opportunity for human response while recognizing the rapid increase in risk associated with prolonged exposure to high temperatures inside a stationary vehicle. All timing logic was implemented using non-blocking software mechanisms to maintain system responsiveness during escalation.

Vision subsystem software development

The vision subsystem was implemented as an independent processing layer within the SmartGuard system to enhance situational awareness, support human presence detection, and provide continuous visual monitoring of the bus interior. Due to the computational demands of image acquisition, processing, and streaming, all vision-related tasks were executed on a Raspberry Pi platform operating separately from the main embedded control unit. This architectural separation ensured that vision processing did not interfere with the deterministic and safety-critical control logic executed by the Arduino MKR NB 1500.

While the vision subsystem processes image data independently, it does not execute safety actions directly. All safety decisions and actuation commands remain under the authority of the Arduino-based control system. The vision subsystem functions as an intelligent data provider that communicates detection results to the embedded control layer, where final decisions are made in coordination with sensor-based detection and system state logic.

The vision subsystem serves three primary functions:

- Processing visual data to detect human presence or motion.
- Supplying detection information to the Arduino control system under defined conditions.
- Providing continuous live video streaming for remote monitoring through the dashboard.

Two cameras were integrated into the vision subsystem; each assigned a distinct role and decision influence level within the overall system.

Vision System Architecture and General Operation

Both cameras continuously capture video frames that are processed locally on the Raspberry Pi using computer vision algorithms tailored to their specific tasks. Image processing and streaming operations are executed independently from the embedded control system, allowing the vision subsystem to function continuously regardless of the Arduino's operational state.

Detection results generated by the vision subsystem are transmitted to the Arduino using structured messaging. These messages include detection flags and timing information but do not trigger actions autonomously. The Arduino evaluates incoming vision data in combination with system state information (such as engine status) and physical sensor outputs before determining whether safety actions should be activated.

This architecture ensures a clear separation between visual data interpretation and safety decision execution, which is a fundamental requirement in safety-critical system design.

Camera 1: Face Recognition Camera

The first camera was dedicated to face recognition–based human detection. This camera continuously processes video frames on the Raspberry Pi to identify the presence of a human face inside the bus. All face recognition processing is performed independently within the vision subsystem.

When Camera 1 detects a human face, the detection information is transmitted to the Arduino control unit. If this detection occurs while the bus engine is in the OFF state, the Arduino treats the face detection as sufficient evidence of human presence and initiates the predefined safety action sequence. In this case, no additional sensor confirmation is required to activate the alert escalation stages.

It is important to note that although Camera 1 has high decision influence, it does not activate actions on its own. Instead, it provides validated detection data that enables the Arduino system to trigger the three-stage safety response in accordance with the system logic. This design reflects the high reliability associated with explicit facial detection while maintaining centralized control of safety actions within the embedded system.

Camera 2: Motion Detection Camera

The second camera was dedicated to motion detection and general interior activity monitoring. Like Camera 1, all motion detection processing is executed independently on the Raspberry Pi. However, due to the higher susceptibility of motion detection to false triggers caused by lighting changes, shadows, or vibrations, Camera 2 was intentionally designed with restricted decision authority.

When motion is detected by Camera 2, the detection data is transmitted to the Arduino control unit. Unlike face recognition, motion detection alone is not sufficient to activate safety actions. The Arduino places the system in a waiting or verification state and monitors additional sensor inputs.

Safety actions are only activated if motion detection from Camera 2 is supported by at least one independent physical sensor, such as the mm Wave radar, thermal sensor, load cell, or passenger counting sensors. This requirement ensures that motion-based detections are corroborated by non-visual sensing layers before escalation occurs.

This dependency-based logic prevents false alerts while allowing Camera 2 to strengthen detection confidence when combined with sensor evidence.

Live Video Streaming Functionality

In addition to detection processing, both cameras continuously provide live video streams that are transmitted to the monitoring dashboard. Live streaming is independent of detection logic and remains active regardless of whether the system is in monitoring mode or alert mode.

The live video feed enables authorized users to remotely observe the bus interior, visually verify alerts, and assess system status in real time. Streaming functionality does not influence safety decisions directly and is designed solely to enhance transparency and situational awareness.

Integration and Safety Control Considerations

Communication between the vision subsystem and the Arduino control unit is implemented using structured messaging with timing validation to ensure data integrity. If vision data become unavailable, delayed, or inconsistent, the Arduino continues to operate using sensor-based detection alone, ensuring that loss of vision functionality does not compromise core safety behavior.

Overall, the vision subsystem was designed to operate independently in processing while remaining tightly coordinated with the embedded control system for safety decision-making. By assigning different authority levels to the two cameras and enforcing sensor-supported validation where required, the SmartGuard system achieves a balanced, reliable, and safety-oriented integration of vision-based intelligence.

3.6.3 Hardware Implementation and Physical Prototyping

Hardware implementation focused on transforming the electronic system design into a physically robust, reliable, and repeatable prototype suitable for integration, testing, and demonstration in a school bus like environment. This stage addressed not only electrical connectivity but also mechanical stability, component protection, and practical deployment considerations. Emphasis was placed on reducing wiring complexity, improving signal reliability, and ensuring consistent sensor positioning, all of which are critical for safety-oriented embedded systems.

The hardware implementation process was divided into two main activities: PCB planning and wiring architecture design, and physical prototype fabrication, including enclosure and sensor mounting solutions.

PCB Design and Wiring Architecture

As the number of sensors, actuators, and communication interfaces increased, it became necessary to move beyond ad hoc breadboard wiring and adopt a more structured hardware interconnection approach. PCB planning was therefore incorporated to improve electrical reliability, reduce noise, and simplify system assembly and maintenance.

The PCB design process began with a detailed analysis of the system's electrical and communication requirements. All components were reviewed to determine voltage levels, current consumption, communication protocols, and pin dependencies. This analysis informed the allocation of microcontroller pins and the routing strategy used in the PCB layout.

Power distribution was another critical aspect of PCB design. The system required both 5 V and 3.3 V power rails to support different sensors and processing units. Voltage regulation and grounding strategies were planned to ensure stable operation and prevent voltage drops or ground loops that could affect sensor accuracy.

To enhance maintainability and scalability, connector-based interfaces were used instead of direct soldered wiring for sensors and actuators. This allowed individual components to be replaced, repositioned, or upgraded without redesigning the entire board.

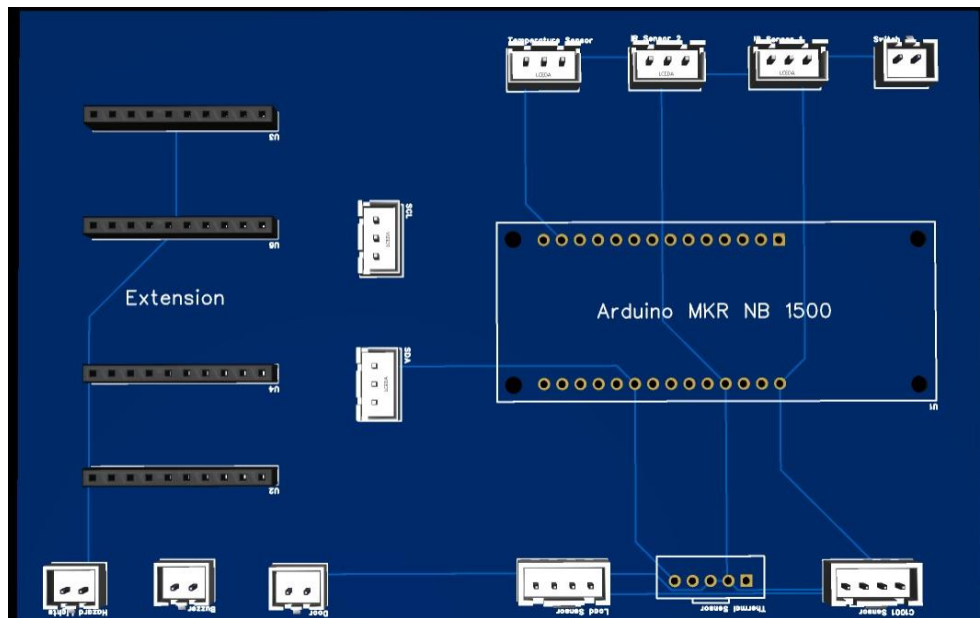


Figure 23: PCB Design

Physical Prototype Fabrication and Enclosure Design

In parallel with electronic hardware planning, physical prototyping was undertaken to house, protect, and stabilize the SmartGuard system components. The objective was to create a mechanically reliable prototype that reflects real-world installation constraints while remaining flexible for iterative development.

A custom enclosure was designed and fabricated using 3D printing technology. The enclosure served multiple purposes: protecting electronic components from accidental contact, organizing internal wiring, and providing structural support for the main processing units. Ventilation openings were incorporated into the design to prevent heat accumulation, particularly around the Arduino and Raspberry Pi, which operate continuously during system operation.

In addition to the main enclosure, custom mounts and holders were designed for individual sensors. These enclosures ensured that each sensor was positioned correctly relative to its intended sensing area and remained fixed during operation.

Mechanical stability was treated as an integral part of system reliability. Loose or unstable sensor placement can lead to inconsistent readings and false detections, particularly in systems that rely on precise thresholds or pattern recognition. By using rigid mounts and enclosure-based fixation, the prototype achieved consistent sensor behavior across repeated runs.



Figure 24: SmartGuard Prototype

Methodological Outcome of Hardware Implementation

The hardware implementation and physical prototyping stage resulted in a structurally stable, electrically reliable SmartGuard prototype that integrates all sensing, processing, and actuation components in a coherent physical form. The combination of PCB planning and custom mechanical fabrication ensured repeatability, reduced wiring-related faults, and improved overall system robustness.

These outcomes established a solid hardware foundation for Phase 5: System Integration Approach, where coordinated operation between all subsystems could be evaluated systematically.

3.7 System Integration Approach

The fifth phase of the SmartGuard project focused on integrating the independently implemented hardware and software subsystems into a single, coordinated safety system. The primary objective of this phase was to ensure reliable interaction between sensing, processing, communication, and actuation components in accordance with the system architecture defined in Phase 3.

System integration was carried out using a structured and incremental approach. Subsystems were combined progressively rather than simultaneously to reduce integration complexity and allow controlled verification of subsystem interactions. This strategy enabled early detection of interface mismatches, timing conflicts, and data coordination issues that are not observable during isolated component implementation.

Integration was centered around the Arduino MKR NB 1500 as the primary control authority. All sensors and vision outputs were integrated as structured inputs to the central decision-making logic, ensuring that safety actions were triggered only through validated system states. The vision subsystem, while operating independently at the processing level, was integrated through defined communication interfaces to support coordinated decision-making without bypassing the embedded control logic.

Communication and actuation subsystems were integrated after core control coordination was established. This ensured that external notifications and physical safety actions were

executed deterministically and independently of network availability. Actuation pathways were integrated last to prevent unintended activation during transitional system states. Initialization and runtime coordination were refined during integration to ensure consistent startup behavior and stable operation. Subsystem availability and communication health were monitored to support coordinated system performance. The outcome of this phase was a fully integrated SmartGuard system in which all subsystems operated cohesively under centralized control.

3.8 Dashboard Development

The sixth phase of the SmartGuard project focused on the design and development of the system monitoring dashboard. The primary objective of this phase was to provide a centralized interface for real-time visualization, system supervision, and situational awareness, enabling authorized users to monitor system status and safety conditions remotely.

Dashboard development was driven directly by the functional and safety requirements defined in Phase 2 and the system architecture established in Phase 3. Rather than being treated as a standalone application, the dashboard was designed as an integral component of the SmartGuard system, closely coupled with sensing, processing, communication, and alert subsystems.

The dashboard design process began by identifying the information required for effective monitoring and decision support. Key data elements included sensor detection status, environmental conditions, passenger presence indicators, system mode (engine ON/OFF), alert stage status, and bus location information. In addition, live video feeds from the vision subsystem were incorporated to support visual verification and remote supervision.

Dashboard functionality was implemented to support real-time data updates, clear status indication, and intuitive presentation of safety-related information. Data received from the SmartGuard system were organized into structured visual elements to ensure that critical information could be quickly interpreted during normal operation and emergency situations.

Special attention was given to prioritizing safety alerts and presenting system states in a clear and unambiguous manner.

Integration with the communication subsystem was a key aspect of this phase. The dashboard was designed to receive system updates, alerts, and location data through the defined communication channels, ensuring synchronization between the physical system and the monitoring interface. The dashboard logic was implemented to reflect system state changes promptly while maintaining consistent operation under varying network conditions. The development process followed an iterative approach, allowing dashboard components to be refined based on system integration progress and evolving monitoring requirements. This ensured compatibility with embedded system outputs, vision subsystem data, and alert mechanisms as they were integrated in Phase 5.

The outcome of this phase was a functional monitoring dashboard capable of displaying real-time system information, alerts, and video feeds in a consolidated interface. This dashboard formed a critical part of the SmartGuard system by enhancing operational transparency and supporting remote supervision, and it was subsequently included in the testing and validation strategy.

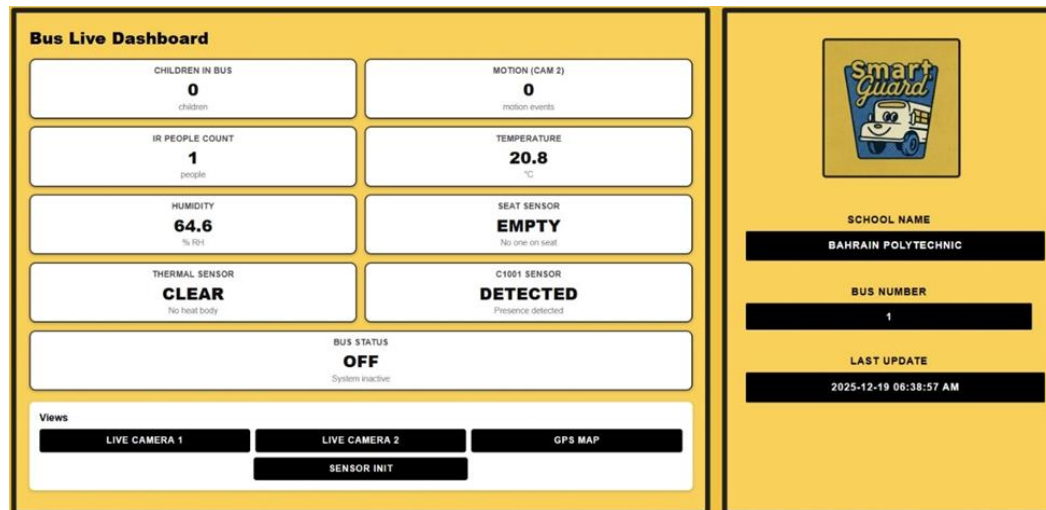
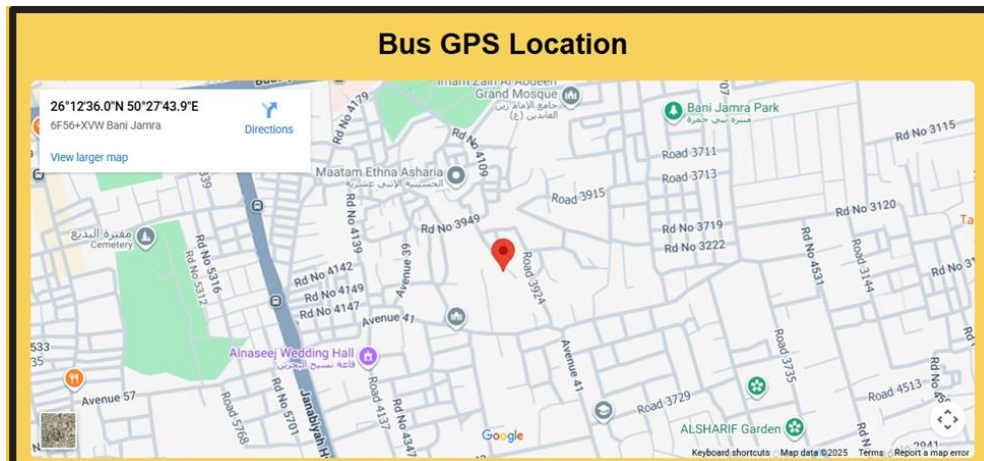


Figure 25: SmartGuard Dashboard Design



3.9 Testing and Validation Strategy

The seventh phase of the SmartGuard project focused on defining a structured testing and validation strategy to evaluate system behavior against the requirements specified in Phase 2. The primary objective of this phase was to establish clear procedures, scenarios, and acceptance criteria that would be used to verify correct system operation and validate that the implemented solution met its intended safety objectives.

The testing and validation strategy was developed in alignment with the hybrid V-Model adopted for the project, ensuring traceability between system requirements, design decisions, and planned verification activities. Testing activities were categorized into multiple levels to address different aspects of system behavior, ranging from individual components to full system operation.

Unit-level verification was planned to assess the basic functionality of individual sensors, embedded software modules, vision-processing components, and dashboard data handling mechanisms. This level of testing aims to confirm correct data acquisition, communication, and response behavior for each component in isolation before higher-level integration.

Integration-level verification was defined to evaluate interactions between subsystems, including sensing, processing, communication, actuation, and dashboard components. Test scenarios at this level focused on ensuring consistent data flow, synchronized system states,

and correct execution of decision-making logic when multiple subsystems operated simultaneously.

System-level validation was planned to assess overall SmartGuard behavior under representative operational scenarios. These scenarios included normal bus operation, post-trip engine shutdown conditions, and situations requiring alert escalation. Validation focused on confirming that the system transitioned correctly between operational modes, applied multi-sensor confirmation rules as intended, and initiated appropriate safety actions according to defined logic and timing requirements.

Dashboard-related validation was included as part of system-level testing to ensure accurate visualization of system status, sensor readings, alerts, and video feeds. This ensured that the monitoring interface reliably reflected the actual state of the SmartGuard system during both normal operation and safety-critical conditions.

Acceptance criteria were defined based on functional, non-functional, and safety requirements to determine whether the system's behavior met the specified objectives. These criteria provided a clear basis for evaluating correctness, reliability, and consistency of system responses.

The execution of the defined testing procedures, collection of test data, and analysis of system performance are presented and discussed in Chapter 4: Results and Discussion. This separation ensured that Chapter 3 focused on methodology and process definition, while Chapter 4 addressed experimental outcomes and performance evaluation.

3.10 Constraints, Assumptions, and Limitations

The design and implementation of the SmartGuard system were subject to several constraints, assumptions, and limitations that influenced the scope, development process, and evaluation of the project. These factors are presented to provide a transparent and realistic assessment of the methodology and to clarify the conditions under which the proposed solution was developed.

One primary constraint was the limited project duration and available resources, as the work was conducted within the timeframe of an academic final-year project. Consequently,

SmartGuard was developed as a functional prototype rather than a fully commercialized product. Design decisions, component selection, and system complexity were therefore balanced against time, budget, and feasibility considerations.

The project assumed the availability of stable power supply from the vehicle electrical system to support continuous operation of sensors, processing units, and communication modules. It was also assumed that all hardware components were installed correctly and maintained in their intended positions. Improper installation, physical obstruction, or damage to sensors could affect detection accuracy and overall system behavior.

Although a multi-sensor detection strategy was adopted to enhance reliability and reduce dependence on individual components, system performance remained partially dependent on correct sensor operation and calibration. Environmental factors such as extreme heat, electrical noise, vibration, and lighting conditions could influence sensor readings, particularly for infrared and vision-based components.

Communication-related limitations were also identified. Remote monitoring and alert transmission relied on cellular network availability, which may vary depending on geographic location and network coverage. While local safety actions were designed to operate independently of network connectivity, delays or interruptions in external communication could affect real-time dashboard updates or remote notifications.

Testing and validation activities were conducted under control and representative conditions due to limited access to real operational school buses and long-term deployment environments. As a result, system behavior under extended real-world usage, varying user behavior, and prolonged environmental exposure may differ from observed test conditions. These factors represent opportunities for further evaluation and system refinement in future work.

Despite these constraints and limitations, the methodology adopted ensured that safety-critical functionality, redundancy, and autonomous operation were prioritized throughout the project. The identified limitations do not undermine the validity of the proposed solution but instead highlight areas for future enhancement, scalability, and real-world deployment considerations.

4. RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents and discusses the results obtained from the implementation and testing of the SmartGuard system. The purpose of this chapter is to evaluate the performance of the developed prototype under representative operating conditions and to assess how effectively the system meets the project objectives defined in earlier chapters.

The results presented in this chapter focus on experimental testing outcomes related to sensor performance, camera-based detection, passenger counting, environmental monitoring, automated emergency response, and communication reliability. Emphasis is placed on analyzing system behavior rather than describing design or implementation details. The discussion examines how the integration of multiple sensing technologies and vision-based intelligence improves detection reliability, reduces false alarms, and enables autonomous safety responses. In addition, the applicability of the SmartGuard system within the Bahraini school transportation context is considered, alongside system limitations identified during testing. The findings presented in this chapter form the basis for the conclusions and recommendations.

4.2 Test Scenarios and Experimental Conditions

To ensure that the evaluation of the SmartGuard system reflects realistic operational conditions, testing was conducted using a set of controlled experimental scenarios designed to simulate typical and critical situations encountered in school bus environments. These scenarios were derived directly from the system requirements and validation strategy defined previously

4.2.1 Test Environment

All experiments and tests were performed in a simulated bus-like environment that replicates the interior conditions of a school bus. Sensors and cameras were installed in positions representative of real-world deployment, including seating areas, entry points, and interior

monitoring zones. A toggle switch was used to simulate the bus ignition state, allowing controlled testing of system behavior when the engine is ON and OFF.

Environmental conditions such as lighting and ambient temperature were varied during testing to evaluate system robustness under different operating circumstances. Testing included normal lighting, reduced lighting conditions, and elevated temperatures to reflect Bahrain’s climatic conditions. The system was powered continuously to assess stability during extended operation.

4.2.2 Test Scenarios

A comprehensive set of test scenarios was defined to evaluate the SmartGuard system under both normal operating conditions and safety-critical situations. These scenarios were designed to verify correct system behavior across different sensor combinations, camera inputs, ignition states, and child positions inside the bus. The scenarios specifically address detection reliability, false alarm prevention, and emergency response logic.

Table 10: SmartGuard Test Scenarios

Scenario ID	Description	Purpose
T1	Single child seated with minimal movement	Validate motionless human detection using mm Wave, thermal, and load sensors
T2	Child sleeping on seat	Verify detection of low-activity occupants and confirm mm Wave, thermal, and load sensor reliability
T3	Multiple passengers entering and exiting	Evaluate passenger counting accuracy using IR sensors and camera-based counting
T4	Single sensor false trigger (noise or transient detection)	Test false alarm prevention and multi-sensor validation logic
T5	Multi-sensor human detection after engine OFF	Validate correct emergency trigger conditions and decision logic

T6	Prolonged unattended presence after engine OFF	Assess time-based alert escalation behavior
T7	Engine ON with multi-sensor human detection	Confirm that no emergency actions are triggered when ignition is ON
T8	Camera-only detection scenarios (Cam1 and Cam2)	Evaluate camera authority levels and interaction with sensor confirmation logic
T9	Child lying on the bus floor	Verify detection capability for non-seated, low-profile human presence

Each scenario was executed repeatedly under controlled conditions. Sensor outputs, camera detections, system states, and alert behavior were logged and analyzed to ensure consistency and correctness.

4.3 Sensor Performance Results

This section presents and analyses the performance of the SmartGuard sensing subsystem based on the experimental scenarios defined. The sensing subsystem forms the foundation of the system's safety logic by providing reliable human presence detection, passenger movement tracking, and environmental awareness. Results are discussed with respect to detection accuracy, reliability under different operating conditions, and contribution to false alarm prevention.

Testing focuses on evaluating individual sensor behavior as well as the effectiveness of multi-sensor validation under safety-critical conditions.



Figure 27: Testing Results

4.3.1 Human Presence Detection Results

Human presence detection is the primary safety function of the SmartGuard system. Detection performance was evaluated using the C1001 mm Wave radar sensor, the AMG8833 thermal imaging sensor, and HX711 load sensors. These sensors were tested individually and in combination across several scenarios.

The C1001 mm Wave sensor demonstrated high sensitivity to human presence, including scenarios involving minimal or no movement. This capability proved particularly important for detecting children who were sleeping or remaining motionless. During early calibration stages, occasional false detections were observed due to environmental reflections and sensor sensitivity. These were eliminated by refining detection thresholds and restricting sensor influence to validated operating states, confirming stable and repeatable performance during final testing.

The AMG8833 thermal sensor consistently detected body heat signatures in all relevant scenarios, including low-light conditions and the floor-lying child scenario. Thermal detection remained unaffected by lighting variations and camera field-of-view limitations, making it a highly reliable confirmation source. Its contribution was especially critical in scenarios where visual confirmation was limited or unavailable.

The HX711 load sensors, installed beneath seating positions, produced stable and repeatable results when a child was seated. While load sensors alone cannot distinguish between a human and a heavy object, their integration into the multi-sensor validation logic significantly increased detection confidence when combined with radar or thermal confirmation.

Table 11: Human Presence Sensor Performance Summary

Sensor Type	Detection Method	Performance Outcome	Key Observations
C1001 mm Wave	Presence and micro-movement	High accuracy after calibration	Effective for motionless occupants
AMG8833 Thermal	Body heat detection	Very high reliability	Independent of lighting conditions

HX711 Load Cell	Seat pressure sensing	Stable and consistent	Requires correct calibration and placement
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When integrated using the multi-sensor validation logic, the system reliably detected human presence with high confidence. Emergency actions were triggered only when the ignition state was OFF and at least two independent detection sources confirmed presence. In contrast, during scenarios when engine ON, no emergency actions were initiated despite multiple sensors detecting occupants, demonstrating correct ignition-gated logic and effective false alarm prevention.

The results confirm that multi-sensor redundancy significantly improves reliability and safety compared to single-sensor approaches, particularly in safety-critical applications.

4.3.2 Passenger Counting Results

Passenger counting was evaluated using infrared (IR) sensors and camera-based counting as part of several scenarios. The IR sensors successfully detected entry and exit events under normal operating conditions, providing reliable passenger count tracking when individuals passed sequentially through the bus doorway.

Minor inaccuracies were observed when multiple passengers entered or exited simultaneously, which is a known limitation of beam-based counting systems. To address this, the SmartGuard system does not rely on passenger count as a standalone safety trigger. Instead, passenger count data is used as a supportive confirmation source within the multi-sensor validation framework.

Camera-based passenger counting provided additional redundancy by tracking registered individuals through facial recognition. This approach proved effective in scenarios involving multiple simultaneous entries, where visual identification remained reliable.

By requiring passenger count confirmation to be supported by at least one additional sensor before triggering emergency actions, the system-maintained safety reliability without increasing false alarms. These results demonstrate that passenger counting enhances contextual awareness while preserving conservative safety decision-making.

4.3.3 Environmental Monitoring Results

The system uses a DHT22 temperature and humidity sensor to continually measure ambient temperatures within the bus to analyze the conditions inside the bus and give supporting data for safety review.

Testing verified that the DHT22 sensor produced reliable and accurate temperature and humidity readings under various circumstances. The sensor provided reliable data needed for continuous monitoring and responded fast to temperature changes. these measurements are crucial In Bahrain environment, where high ambient temperatures greatly increase the danger of suffocation.

Although environmental parameters alone do not trigger emergency actions, their integration strengthens situational awareness and supports informed safety response logic.

4.3.4 Discussion of Sensor Subsystem Performance

Overall, the sensing subsystem demonstrated reliable and consistent performance across all defined scenarios, including edge cases such as a child lying on the bus floor. The results confirm that no single sensor is sufficient to guarantee safety under all conditions. Instead, the combination of radar, thermal, load, and passenger counting sensors provides robust detection coverage and significantly reduces false negatives.

The experimental results validate the design choice of multi-sensor confirmation rather than single sensor triggering. This approach ensures dependable operation under realistic conditions and aligns with safety-critical engineering best practices.

4.4 Camera System Results

This section presents and analyses the performance of the SmartGuard vision subsystem based on the experimental scenarios defined. The camera system was implemented to complement the sensing subsystem by providing visual confirmation, passenger identification, and real-time monitoring of the bus interior. Rather than acting as a standalone

safety mechanism, camera-based detection was evaluated as part of the overall multi-layered validation strategy.

Two cameras were deployed on a Raspberry Pi platform; each assigned a distinct role and decision authority level. Experimental testing focuses on detection accuracy, robustness under different conditions, and correct interaction with sensor-based logic.

4.4.1 Face Recognition Results

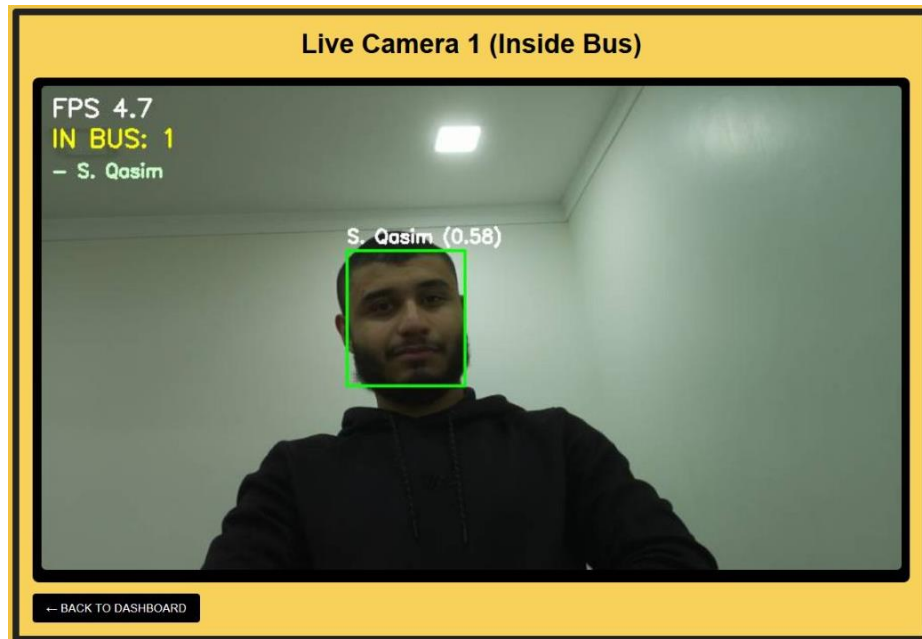


Figure 28: Camera-1 Result

Camera 1 was dedicated to facial recognition and registered passenger tracking. Testing was conducted under several scenarios, with emphasis on evaluating identification accuracy and decision reliability after engine shutdown.

Experimental results showed that the facial recognition algorithm successfully identified registered individuals under controlled conditions when faces were clearly visible, adequately illuminated, and captured within an appropriate distance range. The system reliably differentiated between registered and unregistered individuals, classifying unknown faces correctly.

Recognition accuracy decreased when faces were partially occluded, very poorly illuminated, or captured at extreme angles. These limitations are consistent with known constraints of vision-based recognition systems and were anticipated during system design.

Despite these constraints, Camera 1 demonstrated sufficiently high reliability to be granted elevated decision authority. When a face (registered or unregistered) was detected while the engine was OFF, the system correctly initiated emergency actions without requiring additional sensor confirmation. This design choice was validated during testing and did not result in false emergency triggers due to the high confidence associated with explicit facial detection.

The results confirm that face recognition provides a strong and reliable presence indicator, particularly when combined with ignition state verification.

4.4.2 Visual Passenger Counting Results

In addition to facial recognition, Camera 1 was used to implement a visual passenger counting mechanism. This feature was evaluated as a redundant confirmation layer alongside the IR-based counting system.

The camera-based counting system successfully tracked registered individuals entering and exiting the bus by identifying repeated facial appearances. During testing, the system accurately incremented and decremented the passenger count even when multiple individuals passed through the camera field of view simultaneously.

Unlike IR beam-based counting, visual counting was less affected by simultaneous movement, making it a valuable redundancy mechanism. When the camera-based passenger count remained greater than zero after engine shutdown, the system correctly initiated emergency actions without waiting for additional sensor confirmation. This behaviour was validated experimentally and did not result in false alarms.

The inclusion of visual passenger counting significantly enhanced system robustness by reducing reliance on a single counting method and providing persistent awareness of passenger presence even when individuals were no longer visible in the camera frame.

4.4.3 Motion Detection and Live Streaming Results

Camera 2 was configured for motion detection and continuous live video streaming of the bus interior. Its performance was evaluated under different scenarios, focusing on detection reliability, false trigger susceptibility, and streaming stability.

Motion detection reliably identified human movement within the seating area and central aisle under stable camera placement. Detection performance remained consistent across varying lighting conditions, including reduced lighting environments. However, motion-based detection was sensitive to camera vibration and environmental movement, which occasionally produced transient false detections during early testing.

To mitigate this, Camera 2 was intentionally assigned restricted decision authority. Motion detection alone did not trigger emergency actions. Instead, motion detection required confirmation from at least one independent sensor (mm Wave radar, thermal sensor, load sensor, or passenger count) before escalation was allowed. Experimental results confirmed that this dependency-based logic effectively prevented false alarms while preserving detection sensitivity.

Live video streaming was maintained with low latency and stable frame rates throughout testing. The video feed remained available during all system states, enabling remote visual verification without influencing safety decision logic.

4.4.4 Camera-Based Detection in Edge Scenarios

Special attention was given to scenario T9, where a child was lying on the bus floor rather than seated. In this case, load sensors were not activated, and visual angles were less favorable. Camera-based detection alone was not always sufficient to confirm presence. However, when combined with mm Wave and thermal sensor confirmation, the system consistently detected human presence and initiated appropriate actions.

This result highlights the importance of sensor–vision complementarity and validates the system’s design philosophy of avoiding reliance on any single detection modality.

4.4.5 Discussion of Vision Subsystem Performance

The vision subsystem performed effectively as a complementary detection and monitoring layer. Assigning different authority levels to the two cameras proved critical in balancing detection sensitivity with false alarm prevention. High-confidence facial detection was allowed to act independently, while motion-based detection was conservatively constrained by sensor confirmation.

Overall, the results demonstrate that camera-based intelligence significantly enhances situational awareness and system confidence without compromising safety reliability. When integrated within the multi-sensor validation framework, the vision subsystem contributes meaningfully to accurate detection, redundancy, and real-time monitoring.

4.5 Emergency Response and Automation Results

This section presents and analyses the performance of the SmartGuard system's automated emergency response mechanism. The emergency subsystem is responsible for initiating safety actions when a verified human presence is detected inside the bus after engine shuts down. Testing focuses on evaluating trigger accuracy, response timing, escalation behavior, and reliability under different detection scenarios.

4.5.1 Emergency Trigger Conditions

Emergency actions in the SmartGuard system are governed by a strict set of trigger conditions designed to prevent false alarms while ensuring timely response in genuine risk situations. Experimental testing confirmed that emergency activation occurred only when the following conditions were satisfied:

- The bus ignition state was OFF
- Human presence was confirmed by: At least two independent detection sources, human detected by camera 1

Table 12: Emergency Trigger Condition Validation

Condition ID	Ignition State	Active Detection Sources	Emergency Action
C1	ON	Any single or multiple sensors	No action
C2	ON	Any camera	No action
C3	OFF	Single sensor only	No action
C4	OFF	Camera 2 only	No action
C5	OFF	Any 2 or more sensors	Actions activated
C6	OFF	Sensor + Camera 2	Actions activated
C7	OFF	IR passenger count > 0 + Sensor or Camera 2	Actions activated
C8	OFF	Camera 1 detects face	Actions activated
C9	OFF	Camera 1 count > 0	Actions activated

During repeated testing, the system consistently adhered to these conditions. No emergency actions were triggered during engine ON scenarios, even when multiple sensors detected occupants, confirming correct ignition-gated logic. This behavior is critical for preventing unnecessary alarms during normal bus operations.

4.5.2 Automated Safety Actions and Escalation Behavior

Once emergency conditions were verified, the SmartGuard system executed a staged response strategy designed to balance rapid intervention with proportional escalation. Due to prototype limitations, physical actions were simulated using indicator LEDs representing real-world actuations.

The implemented escalation stages are summarized below:

Stage 1 (Immediate Response): Triggered immediately after emergency verification.

- SMS alert sent to the bus driver
- System enters emergency monitoring state

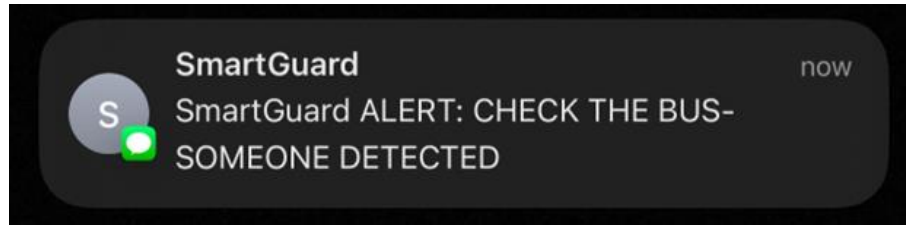


Figure 29: Actions-Stage-1

Stage 2: Triggered if no response is detected after a predefined time interval.

- Horn activation, hazard lights activated, and bus door opened
- Telegram alert sent to parents including bus location

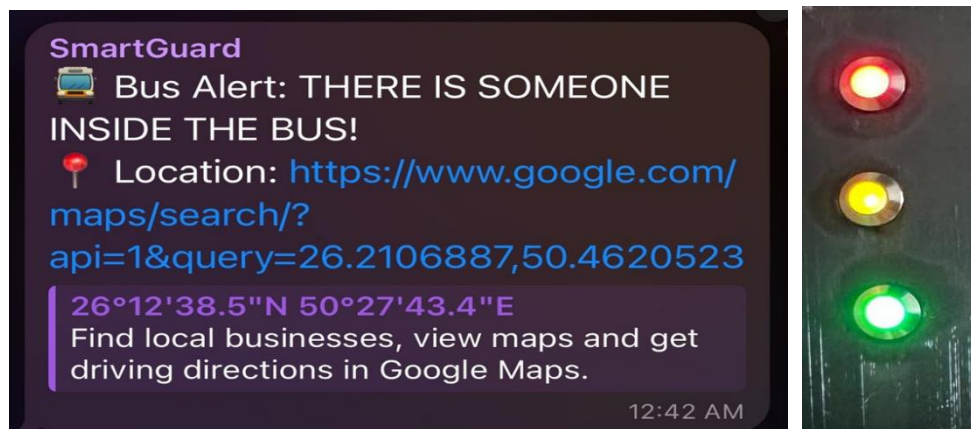


Figure 30: Actions-Stage-2

Stage 3 (Critical Escalation): Triggered if the hazardous condition persists.

- Automated phone call to emergency contact or authority

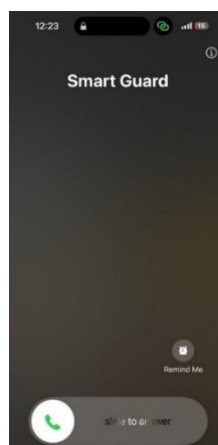


Figure 31: Actions-Stage-3

Experimental results confirmed that all stages were executed in the correct sequence and without timing conflicts. Transition between stages occurred deterministically and was not affected by sensor noise or transient detections.

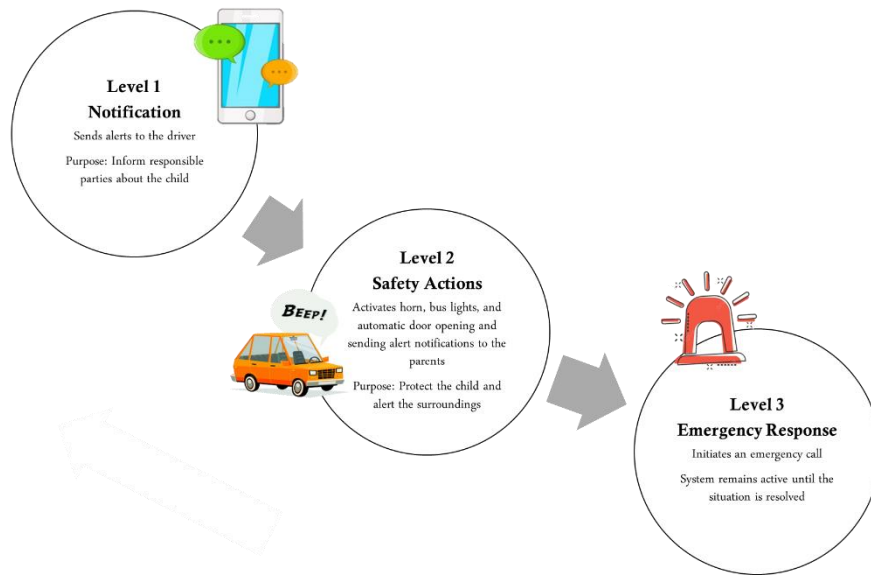


Figure 32: SmartGuard's Action stages

4.5.3 Response Timing and System Reliability

Response timing was evaluated across the scenarios to ensure that safety actions were initiated without unnecessary delay. The system demonstrated immediate Stage 1 activation once trigger conditions were met, followed by accurate timing of escalation stages.

Importantly, all escalation logic was implemented using non-blocking software mechanisms. This ensured that sensor monitoring, communication, and decision-making remained active throughout the emergency sequence.

No missed escalations, repeated triggers, or system freezes were observed during testing. The system maintained stable operation even during prolonged emergency states, demonstrating reliable real-time behavior suitable for safety-critical applications.

4.5.4 False Alarm Prevention Performance

Preventing false alarms is essential for maintaining trust and reliability in automated safety systems. Testing under scenarios involving single-sensor noise, transient detections, and motion-only events confirmed that the SmartGuard system did not trigger emergency actions under invalid conditions.

Emergency actions were not activated in the following cases:

- Engine ON with confirmed human presence
- Single-sensor detections after engine OFF
- Short duration or transient detections
- Motion only detection without sensor confirmation

These results demonstrate that the multi-sensor validation strategy effectively balances sensitivity and reliability. By requiring confirmation from independent sources or high confidence visual detection, the system minimizes false alarms without compromising detection capability.

4.5.5 Discussion of Emergency Automation Performance

The emergency response results confirm that SmartGuard operates as a proactive, autonomous safety system rather than a passive monitoring solution. The staged escalation strategy ensures that human intervention is encouraged early while guaranteeing autonomous action if no response occurs.

The system's ability to correctly distinguish between safe and hazardous conditions, combined with reliable automation and false alarm prevention, demonstrates strong alignment with safety-critical engineering principles. These results validate the effectiveness of SmartGuard's emergency response design in addressing the risk of children being left unattended inside school buses.

4.6 Communication and Notification Results

The results pertaining to the communication and notification capabilities of the SmartGuard system are shown in this section. To guarantee that confirmed emergency situations are

quickly communicated to the appropriate individuals outside the vehicle, effective communication is crucial. Testing concentrated on assessing system behavior under various network circumstances, alert generating accuracy, and transmission dependability.

4.6.1 Alert Generation and Transmission Performance

The SmartGuard system uses the cellular communication capabilities of the Arduino MKR NB 1500 to transmit alerts when emergency conditions are verified. Alerts were generated only after the system entered an emergency state, ensuring that notifications were not triggered by transient or unverified detections.

During experimental testing, SMS alerts to the bus driver were generated immediately upon Stage 1 activation. Telegram messages to parents, including location information, were successfully transmitted during Stage 2 escalation. Automated call initiation was triggered correctly during Stage 3 escalation when the hazardous condition persisted.

The system demonstrated stable alert behavior, with each emergency event resulting in a single, well-defined notification sequence. No duplicate, repeated, or unintended alerts were observed during testing, confirming correct event-state management.

4.6.2 Communication Latency and Reliability

Alert delivery latency was influenced by cellular network conditions, which varied depending on signal strength and coverage. In areas with strong connectivity, alerts were delivered within acceptable time frames for safety intervention. In weaker coverage conditions, minor delays were observed in external notification delivery.

Importantly, local safety actions such as horn activation, hazard lights, and door opening were not dependent on network availability. These actions were executed immediately upon emergency verification, ensuring that system safety was not compromised by communication delays or connectivity loss.

This design decision significantly enhances system reliability, as core safety behavior remains functional even during communication interruptions.

4.6.3 Communication System Limitations

While communication performance was generally reliable, certain limitations were identified during testing. Cellular connectivity depends on network availability, which may vary across geographic locations. Additionally, message delivery times may be affected by network congestion or service provider constraints.

These limitations do not affect the system's ability to detect hazardous conditions or initiate local safety actions. Instead, they primarily impact the timing of remote notifications. Future system enhancements could incorporate redundant communication channels or alternative data transmission methods to further improve reliability.

4.6.4 Discussion of Communication Performance

Overall, the communication subsystem successfully fulfilled its intended role by delivering timely and accurate alerts during verified emergency situations. The integration of communication logic with multi-sensor validation ensured that notifications were meaningful and not excessive.

By decoupling local safety actions from network availability, the SmartGuard system demonstrates a robust and safety-oriented communication design suitable for real-world deployment in school transportation environments.

4.7 Discussion of Results

This section discusses the results by evaluating the SmartGuard system's performance in relation to the project objectives, safety requirements, and real-world operating conditions. The discussion focuses on the effectiveness of the multi-sensor validation strategy, the reliability of automated decision-making, and the practical implications of the results.

4.7.1 Achievement of Project Objectives

One of the primary objectives of the SmartGuard project was to develop an automated system capable of reliably detecting children inside a school bus after engine shutdown and initiating appropriate safety actions. The experimental results demonstrate that this objective was successfully achieved.

Human presence was accurately detected across a wide range of scenarios, including motionless occupants, sleeping children, and edge cases such as a child lying on the bus floor. The use of complementary sensing technologies ensured that detection reliability was maintained even when individual sensors were limited by environmental conditions or physical positioning. This confirms that the system effectively reduces dependence on any single detection method.

Another key objective was the reduction of human error in post-trip safety procedures. The results show that SmartGuard operates autonomously once the ignition is switched off, requiring no manual confirmation from the driver to initiate safety actions. Emergency responses were triggered consistently and only under verified risk conditions, demonstrating that the system functions as a proactive safety mechanism rather than a passive monitoring tool.

The objective of implementing automated, staged emergency responses was also fulfilled. The system reliably executed escalating safety actions in the correct sequence and timing, ensuring that early intervention opportunities were provided while guaranteeing autonomous escalation when no response occurred. Communication objectives were achieved through consistent alert generation and delivery, with local safety actions remaining effective even in the presence of network limitations.

Overall, the experimental results confirm that all core project objectives were met under representative operating conditions.

4.7.2 Effectiveness of the Multi-Sensor Validation Strategy

The results strongly support the effectiveness of the multi-sensor validation approach adopted in SmartGuard. Single-sensor detection was intentionally insufficient to trigger emergency actions, which significantly reduced the likelihood of false alarms caused by noise, environmental interference, or transient events.

By requiring confirmation from at least two independent sensing sources, or a high-confidence visual event, the system demonstrated a balanced trade-off between sensitivity and reliability. This was particularly evident in scenarios involving motionless occupants and children positioned outside standard seating areas, where individual sensors alone might fail.

The inclusion of both physical sensors and vision-based detection created a layered verification framework that enhanced robustness. This design choice aligns with best practices in safety-critical system engineering, where redundancy and diversity of sensing modalities are essential for reliable operation.

4.7.3 Camera Authority and Decision-Making Design

A critical design decision evaluated through testing was the assignment of different authority levels to the two cameras. Granting Camera 1 elevated authority due to the high confidence associated with explicit facial detection proved effective and did not introduce false alarms. Conversely, restricting Camera 2's authority and requiring sensor confirmation for motion-based detection prevented unnecessary escalation.

The results validate this asymmetric camera design, demonstrating that intelligent allocation of decision authority can enhance system reliability without sacrificing responsiveness. This approach ensures that visual intelligence strengthens, rather than destabilizes, the overall safety logic.

4.7.4 System Reliability and False Alarm Prevention

False alarm prevention is essential for maintaining trust in automated safety systems. The results show that SmartGuard consistently avoided emergency activation during invalid conditions, including engine ON states, single-sensor detections, and short-duration events. The absence of false emergency escalations during testing indicates that the system's decision logic is conservative without being unresponsive. This balance is critical for real-world deployment, where frequent false alarms could lead to user desensitization or system disablement.

4.7.5 Limitations and Practical Constraints

Despite the positive results, certain limitations were identified. Passenger counting accuracy may decrease during simultaneous entry or exit events, and facial recognition performance depends on lighting conditions, camera angle, and dataset quality. Additionally, communication latency is influenced by cellular network availability.

However, these limitations do not compromise the system's core safety functionality. The multi-sensor validation framework ensures that detection reliability is preserved even when individual subsystems experience temporary degradation. Furthermore, local safety actions remain independent of network connectivity, ensuring continuous protection.

Testing was conducted in a simulated environment due to practical constraints, which may not fully capture long-term real-world variations such as vibration, prolonged exposure, or user behavior. These factors represent areas for future validation rather than fundamental system weaknesses.

4.7.6 Overall Interpretation

The results demonstrate that SmartGuard functions as a reliable, autonomous, and safety-oriented system capable of addressing the critical risk of children being left unattended inside school buses after turning them OFF. The integration of multiple detection modalities, conservative decision logic, and staged emergency response mechanisms collectively contribute to a robust and practical solution.

The discussion confirms that the system design choices made during development were appropriate and effective, and that the implemented solution aligns well with the intended safety objectives and real-world operational requirements.

4.8 Applicability and Impact in Bahrain

The applicability of the SmartGuard system within the Kingdom of Bahrain is strongly supported by the experimental results presented in this chapter. Bahrain's climatic conditions, characterized by high ambient temperatures and high humidity level for much of the year, significantly increase the risk associated with children being left unattended inside stationary vehicles. The system's reliable detection of stable occupants, combined with rapid emergency escalation, directly addresses this environmental risk.

The results demonstrate that SmartGuard operates effectively under elevated temperature conditions like those experienced in Bahraini school buses. Environmental monitoring results confirmed that internal bus temperatures can reach hazardous levels within a short time frame, reinforcing the need for automated, ignition-gated safety systems. The system's ability

to initiate safety actions immediately after engine shutdown makes it particularly suitable for local school transportation practices.

From an operational perspective, the autonomous nature of SmartGuard reduces reliance on manual post-trip inspections, which are vulnerable to human error due to fatigue, routine disruption, or time pressure. This aligns with documented challenges observed in regional transportation incidents and reflects the need for technology-assisted safety enforcement in Bahrain.

The system's modular design further enhances its suitability for local deployment. Components such as sensors, cameras, and communication modules can be scaled or adapted to different bus sizes and operational requirements. Cellular-based alerting enables communication without reliance on local infrastructure, supporting deployment across both urban and remote areas of Bahrain.

Overall, the results confirm that SmartGuard represents a practical, scalable, and context-appropriate solution capable of enhancing child safety in Bahrain's school transportation sector. The system aligns with national priorities related to public safety, smart infrastructure, and child welfare, and supports broader objectives associated with sustainable and technology-driven public services.

4.9 Skills and Knowledge Development

The development and evaluation of the SmartGuard system required the application of a broad range of technical and professional skills relevant to modern electrical and electronic engineering practice. The results presented in this chapter reflect the successful integration of these skills into a functional, safety-critical system.

4.9.1 Technical Skills Applied

The project developed advanced embedded systems skills, including sensor interfacing, microcontroller programming, real-time decision logic, and safety-critical fault handling through multi-sensor data fusion. Computer vision and AI techniques were applied using facial recognition, motion detection, and real-time video streaming on a Raspberry Pi platform. Practical hardware integration skills were strengthened through PCB planning,

sensor installation, wiring management, and 3D-printed enclosure fabrication, reflecting real-world deployable system development.

4.9.2 Skill Development Relevance in Bahrain

The abilities displayed in this project closely relate to Bahrain's increasing emphasis on digital transformation, intelligent transportation solutions, and smart systems. IoT system, automation, embedded intelligence, and safety-critical engineering competencies are in line with national development plans and industry demands.

Engineering students may get hands-on experience utilizing technology to solve actual social challenges through initiatives like SmartGuard. The findings demonstrate how academic education and applied engineering approaches may help find answers to regional problems while preparing graduates for jobs in automation, safety systems, and smart infrastructure.

5. CONCLUSION

5.1 Overview

This chapter presents the overall conclusions drawn from the SmartGuard project and summarizes the key findings obtained throughout all stages of the work. It evaluates the extent to which the project objectives were achieved, highlights the technical, social, and practical impacts of the proposed system, and discusses its benefits within the Bahraini context. Finally, the chapter provides recommendations for future work and potential system improvements based on the results and limitations identified during testing and evaluation.

5.2 Summary of Project Outcomes

The SmartGuard project was developed to address the critical safety issue of children being unintentionally left inside school and kindergarten buses after engine shutdown. Through the integration of embedded systems, multi-sensor detection, computer vision, and automated decision-making, the project successfully demonstrated a proactive and autonomous child-safety solution.

The project progressed systematically from problem identification and requirements definition to system architecture design, hardware and software implementation, system integration, and experimental testing. Each phase contributed directly to the development of a functional prototype capable of detecting human presence under a wide range of realistic scenarios.

Experimental results confirmed that the SmartGuard system reliably detects human presence using a combination of mm Wave radar, thermal sensing, load sensing, infrared passenger counting, and camera-based intelligence. The use of multi-sensor validation significantly reduced false alarms while ensuring accurate detection even in challenging cases such as motionless occupants, sleeping children, and children lying on the bus floor.

The automated emergency response mechanism functioned as intended, initiating staged safety actions only when verified risk conditions were present. Communication and alerting mechanisms successfully delivered notifications to responsible parties while ensuring that

core safety actions remained independent of network availability. These outcomes demonstrate that SmartGuard operates as a dependable and safety-oriented system rather than a passive monitoring solution.

5.3 Achievement of Project Objectives

The primary objectives defined at the beginning of the project were successfully achieved. The system demonstrated the ability to prevent child suffocation risks by autonomously detecting human presence after engine shutdown and initiating appropriate safety actions. The implementation of multiple complementary sensors ensured reliable detection and reduced dependence on human supervision, directly addressing the limitations of traditional manual inspection procedures.

The objective of automating safety responses was fulfilled through a structured escalation strategy that balances early human intervention with guaranteed autonomous action when no response occurs. Real-time monitoring, location tracking, and alert communication objectives were also achieved, enabling both local and remote awareness of safety-critical situations.

Overall, the SmartGuard prototype met its functional, safety, and reliability objectives under representative testing conditions, confirming the technical validity and effectiveness of the proposed solution.

5.4 Impact and Benefits for Bahraini Society

The SmartGuard system offers significant benefits within the Bahraini context, where high ambient temperatures and heavy reliance on school bus transportation increase the risks associated with children being left unattended inside vehicles. The system's ability to detect hazardous conditions quickly and operate autonomously aligns directly with local environmental and societal needs.

By reducing dependence on manual post-trip checks, SmartGuard minimizes the risk of human error caused by fatigue, distraction, or routine disruption. This contributes to enhanced child safety, increased parental confidence, and improved accountability within school transportation systems.

The project also supports Bahrain’s broader objectives related to smart infrastructure, digital transformation, and public safety. The use of intelligent sensing, automation, and IoT-based communication reflects modern engineering approaches that can be extended to other safety-critical applications within the country. As such, SmartGuard demonstrates how locally relevant engineering solutions can contribute to safer, smarter, and more resilient public services.

5.5 Limitations of the System

While the SmartGuard system demonstrated strong performance, certain limitations were identified. Testing was conducted in a simulated environment rather than on fully operational school buses, which may not capture all real-world variations such as prolonged vibration, extended deployment periods, and diverse user behaviors.

Some subsystems, such as passenger counting and facial recognition, are influenced by factors including lighting conditions, camera positioning, and dataset size. Communication performance is also dependent on cellular network availability. However, these limitations do not compromise the system’s core safety functionality, as the multi-sensor validation framework ensures reliable detection even when individual components experience reduced performance.

5.6 Recommendations for Future Work

Based on the project’s outcomes and identified limitations, several recommendations are proposed for future development and improvement of the SmartGuard system:

- **Real-World Field Deployment**

Future work should include long-term testing on operational school buses to evaluate system performance under real driving conditions, extended usage, and varying environmental factors.

- **Enhanced Sensor Redundancy**

Additional sensing technologies, such as CO₂ or oxygen sensors, could be integrated to further strengthen detection reliability and environmental awareness.

- **Improved Vision System Accuracy**

Expanding and diversifying facial recognition datasets and optimizing camera placement would improve recognition performance under varied lighting and seating conditions.

- **Communication Redundancy**

Incorporating alternative communication channels, such as Wi-Fi or satellite-based alerts, could enhance reliability in areas with limited cellular coverage.

- **Commercialization and Regulatory Integration**

Future development could focus on system miniaturization, certification, and compliance with transportation safety regulations to support large-scale deployment within Bahrain.

- **Integration with National Safety Platforms**

Linking SmartGuard with centralized school or emergency response systems could further enhance response coordination and accountability.

5.7 Conclusion

To sum up, the SmartGuard project effectively illustrated the planning, execution, and assessment of an intelligent child-safety system for school transportation. The outcomes verify that the suggested approach successfully resolves crucial safety concern via autonomous detection, trustworthy decision-making, and prompt emergency action.

The project shows how engineering solutions may be customized to solve actual social concerns and emphasizes the need for multi-sensor validation, automation, and intelligent monitoring in safety-critical applications. SmartGuard is a significant contribution to enhancing child safety and developing smart transportation systems in Bahrain, and it offers a solid basis for future development, implementation, and study.

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APPENDICES

Appendix A: Sensors C++ Code

C1001

```
DFRobot_HumanDetection hu(&Serial1);
int previousPresence = -1;

bool INIT_C1001 = false;

unsigned long lastC1001Read = 0;
const unsigned long C1001_INTERVAL = 120; // ms

Health H_C1001 = {false, 0};

// Valid-frame tracking
unsigned long lastC1001ValidMs = 0;
const unsigned long C1001_VALID_TIMEOUT_MS = 1200;

// Consecutive fail tracking
uint16_t c1001FailStreak = 0;

// Recovery tracking (kept, but no longer used in runtime)
unsigned long lastC1001ProbeTryMs = 0;
uint8_t c1001RecoverHits = 0;

// Forward
void printC1001NowAfterReady();

static inline void flushSerial1Fast() {
    while (Serial1.available()) Serial1.read();
}

static inline bool c1001PollValid(uint16_t &hpOut) {
    Serial1.setTimeout(15);
    uint16_t hp = hu.smHumanData(hu.eHumanPresence);
    if (hp == 0xFFFF) return false;
```

Figure 33: C1001 Sensor's Code

Load Sensor (HX711)

```
const int HX711_dout = 6;
const int HX711_sck  = 7;
HX711_ADC LoadCell(HX711_dout, HX711_sck);

bool INIT_LOAD = false;

float calibrationValue = 1.0;
static bool personDetected = false;
const float WEIGHT_THRESHOLD = 10000.0;
FlashStorage(loadCellCalStorage, float);

const float WEIGHT_HYSTERESIS = 600.0;
const uint8_t LOAD_ON_COUNT  = 2;
const uint8_t LOAD_OFF_COUNT = 2;

Health H_LOAD = {false, 0};
unsigned long lastLoadGoodMs = 0;
const unsigned long LOAD_TIMEOUT_MS = 1800;

unsigned long lastLoadInitTryMs = 0;
const unsigned long LOAD_REINIT_INTERVAL_MS = 2500;

// ---- Non-blocking runtime init state machine for HX711 ----
enum LoadInitState { LI_IDLE, LI_TRY_START, LI_WAIT_DATA, LI_TARE_RUNNING };
LoadInitState loadInitState = LI_IDLE;

unsigned long loadInitStateMs = 0;
const unsigned long LOAD_TRY_COOLDOWN_MS = 2500;
const unsigned long LOAD_WAIT_DATA_MS   = 350;
const unsigned long LOAD_TARE_MAX_MS    = 1500;
```

Figure 34: Load Sensor's Code

Humidity and Temperature Sensor (DHT22)

```
#define DHTPIN 4
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);

bool INIT_DHT = false;

unsigned long lastDHTRead = 0;
const unsigned long dhtInterval = 60000;

Health H_DHT = {false, 0};
const unsigned long DHT_RUNTIME_TIMEOUT_MS = 150000;

unsigned long lastDhtInitTryMs = 0;
const unsigned long DHT_REINIT_INTERVAL_MS = 60000;
```

Figure 35: DHT22 Sensor's Code

IR Sensor

```
//  
#define IR1_PIN 2  
#define IR2_PIN 3  
int peopleCount = 0;  
  
enum IRState { IR_IDLE, IR1_TRIGGERED, IR2_TRIGGERED };  
IRState irState = IR_IDLE;  
unsigned long stateStartTime = 0;  
  
const unsigned long IR_TIMEOUT = 700;  
  
unsigned long lastIRPollUs = 0;  
const unsigned long IR_POLL_US = 1000;  
const unsigned long IR_DEBOUNCE_US = 6000;  
  
bool prevIR1 = HIGH;  
bool prevIR2 = HIGH;  
unsigned long lastIR1EdgeUs = 0;  
unsigned long lastIR2EdgeUs = 0;  
  
bool INIT_IR = true;
```

Figure 36: IR Sensors' Code

Thermal Sensor (AMG8833)

```
Adafruit_AMG88xx amg;  
float pixels[64];  
bool thermalPerson = false;  
  
bool INIT_AMG = false;  
bool amgOk = false;  
  
unsigned long lastAMGRead = 0;  
const unsigned long AMG_INTERVAL = 500;  
const float BASE_TEMP = 26.0;  
const float DETECT_THRESHOLD = 3.0;  
  
Health H_AMG = {false, 0};  
  
unsigned long lastAmgInitTryMs = 0;  
const unsigned long AMG_REINIT_INTERVAL_MS = 2500;
```

Figure 37: Thermal Sensor's Code

Appendix B: Cameras Code

```
# 🧑 Person detection on Cam2 (motion)
# =====
def setup_person_detector():
    hog = cv2.HOGDescriptor()
    hog.setSVMDetector(cv2.HOGDescriptor_getDefaultPeopleDetector())
    return hog

def setup_motion_subtractor():
    return cv2.createBackgroundSubtractorMOG2(history=500,
                                              varThreshold=16,
                                              detectShadows=True)

def draw_motion_boxes(frame, fgmask, min_area=1500):
    kernel = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (3, 3))
    fg = cv2.morphologyEx(fgmask, cv2.MORPH_OPEN, kernel, iterations=1)
    fg = cv2.dilate(fg, kernel, iterations=2)
    _, fg = cv2.threshold(fg, 200, 255, cv2.THRESH_BINARY)

    contours, _ = cv2.findContours(fg, cv2.RETR_EXTERNAL,
                                   cv2.CHAIN_APPROX_SIMPLE)

    count = 0
    for c in contours:
        if cv2.contourArea(c) < min_area:
            continue
        x, y, w, h = cv2.boundingRect(c)
        cv2.rectangle(frame, (x, y), (x + w, y + h),
                      (0, 165, 255), 2)
        count += 1
    return count

def detect_people_hog(hog, frame, fgmask=None, min_area=1200,
                     scale_target_h=360, win_stride=(8, 8),
                     padding=(8, 8), scale=1.03):
    H, W = frame.shape[:2]
    rois = []

    if fgmask is None:
        rois.append(((0, 0, W, H), frame))
```

Figure 38: Cameras' Code

```

# ----- A) Cam1 face tracking -----
cam1_face_seen = (len(boxes) > 0) # ANY face detected on Cam1
# If you want ONLY recognized faces, use this instead:
# cam1_face_seen = (inside_cnt > 0)

if cam1_face_seen:
    globals()["_last_face_ts"] = now
face_ok = (now - globals()["_last_face_ts"]) <= FACE_HOLD_S

# ----- B) Cam2 motion tracking -----
if motion_cnt2_val > 0:
    globals()["_last_motion_ts"] = now
motion_ok = (now - globals()["_last_motion_ts"]) <= MOTION_HOLD_S

# "Other sensors" confirmation (same as your old list)
# ----- A) Cam1 -----
cam1_any_face = (len(boxes) > 0) # unknown OR known face
buscount_ok = (inside_cnt > 0) # bus counter > 0

cam1_ok = cam1_any_face or buscount_ok

# ----- B) Cam2 -----
arduino_any_ok = (
    (c1001_present is True) or
    (thermal_present is True) or
    (seat_occupied is True) or
    (ir_count > 0)
)
motion_combo_ok = motion_ok and arduino_any_ok

```

Figure 39: Cameras' Code

Appendix C: 3D Design

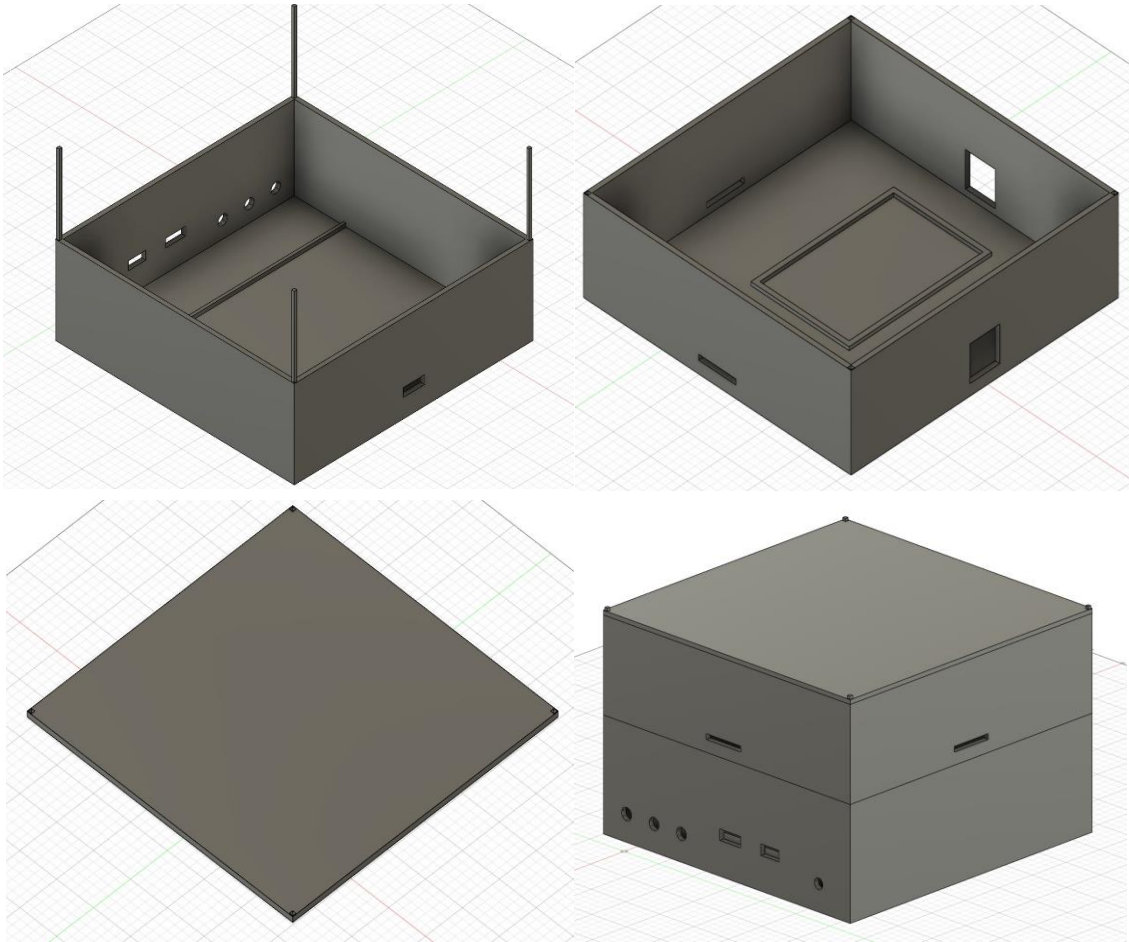


Figure 40: SmartGuard's Prototype 3D Design

Appendix D: PCB

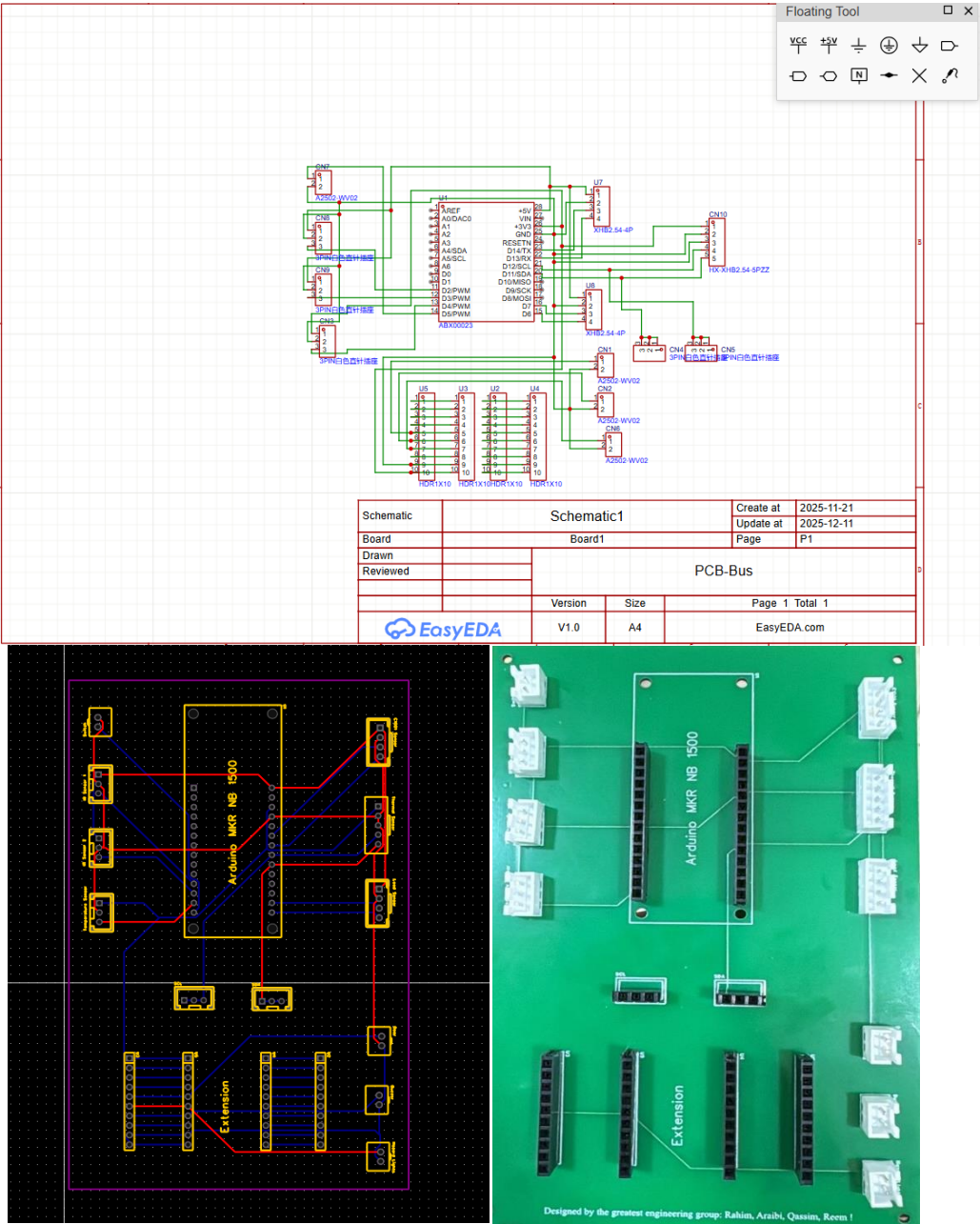


Figure 41: PCB Design Process